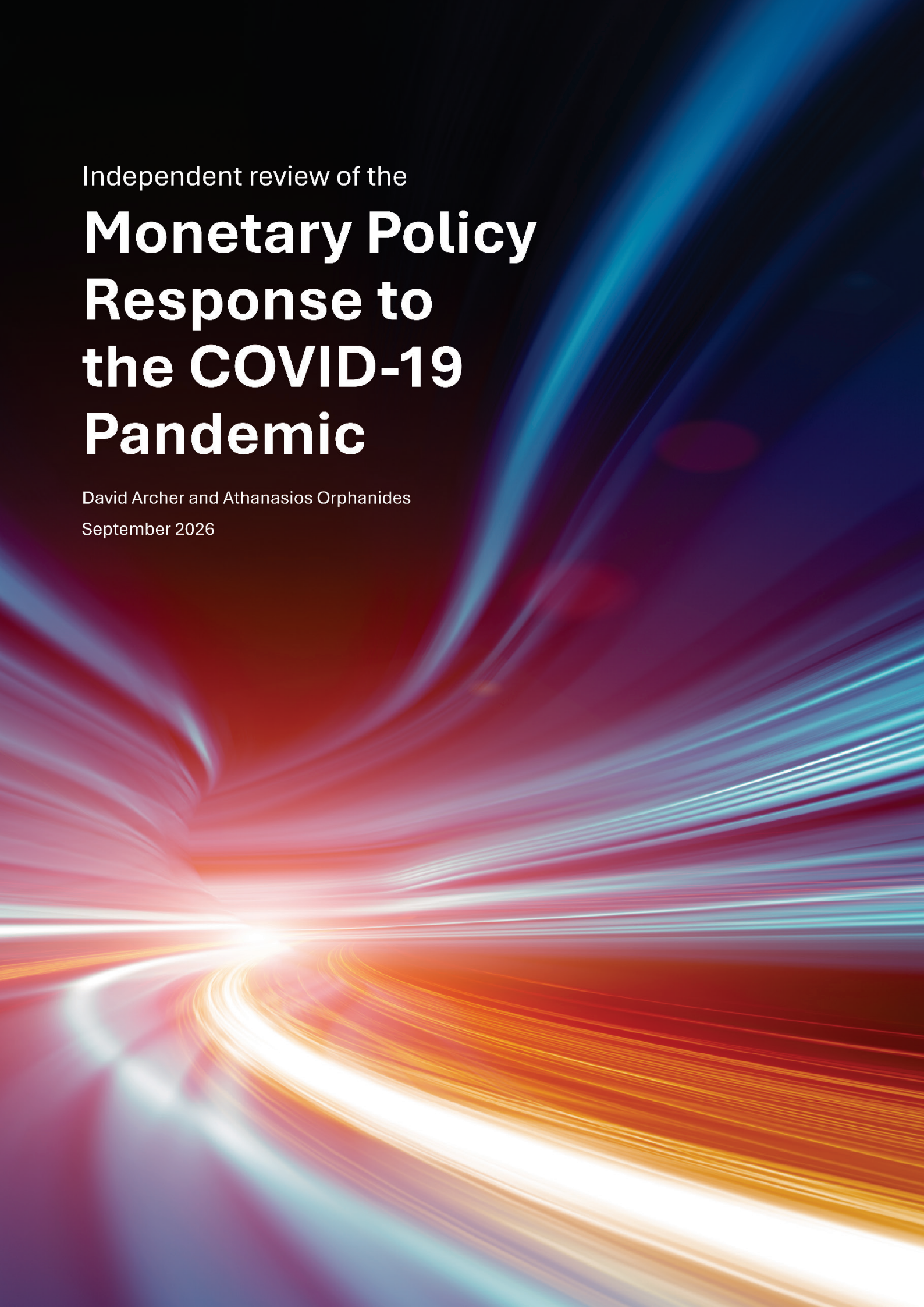


The background of the cover is an abstract composition of light streaks. In the upper right, there are bright blue and white streaks that curve downwards. In the lower left, there are vibrant red and orange streaks that curve upwards. These two sets of streaks converge towards the center, creating a sense of dynamic movement and energy. The overall effect is reminiscent of a high-speed light trail or a stylized representation of data flow.

Independent review of the

Monetary Policy Response to the COVID-19 Pandemic

David Archer and Athanasios Orphanides

September 2026



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Preface and acknowledgements

Towards the end of 2025, the Treasury, on behalf of the Minister of Finance, invited us¹ to assess the performance of New Zealand’s monetary policy response to the COVID-19 pandemic. The stated purpose *is to support accountability and public confidence in the operational independence of monetary policy, and to improve the ongoing effectiveness of monetary policy.*²

The time gap between the review and the 2020 to 2022 period under review created challenges and provided benefits. Understanding decisions in real time—avoiding hindsight—becomes more difficult as time passes, but understanding the consequences of decisions is helped by seeing what happened. Lessons have already been learned, and many gaps have been filled. Accordingly, we concentrate mostly on areas where we believe gaps remain.

The Review was not established as an inquiry under the Inquiries Act 2013. Those who we asked to talk with us had no obligation to do so, yet all agreed willingly and were open and constructive in answering our questions and providing material.³ We express our gratitude to all. Special thanks go to those who were Monetary Policy Committee (MPC) members during the period under review; to the leadership of the Reserve Bank of New Zealand (the Reserve Bank) for a commitment to transparency and learning; to Reserve Bank staff who were generous with their time and willingness to provide historical real-time data and hunt down a wide range of records and documents; to Anella Munro who expertly supported the Review as a consultant; to Sam Thornton for steadfast support and organization; to a large number of former Reserve Bank officials, staff and expert observers who engaged with us.

¹ Biographical information is available in Annex 2.

² The Terms of Reference is available [here](https://www.treasury.govt.nz/publications/information-release/independent-review-monetary-policy) (<https://www.treasury.govt.nz/publications/information-release/independent-review-monetary-policy>).

³ A list of people the reviewers met with is available in Annex 3.

1. Introduction, Conclusions and Recommendations

1.1. Introduction

The Terms of Reference direct the Review to focus on monetary policymaking in the period from 1 January 2020 through 31 December 2022—the COVID-19 period—with two main objectives: support accountability and improve the ongoing effectiveness of monetary policy.

The accountability aspects are to be considered against the policy objectives that applied at the time, having regard to what was known and could reasonably have been known in the circumstances, and to relevant decisions made by comparable central banks. The latter reference point is apt, since the COVID-19 pandemic was global and the monetary policy challenges in different jurisdictions were similar. Rapid curtailment of economic activity called for assertive macroeconomic policy responses everywhere; financial markets came under stress in many places; and the post-pandemic inflation surge was widespread. Policy responses were impressive for their speed, scale and imagination. Yet, unsurprisingly in the circumstances, mistakes were made, not just in New Zealand.

The commonality of the policy successes and failures across several countries directs attention to common features of policy strategy and conduct in uncertain conditions. Strategic or structural weaknesses that are similar and that become exposed in such conditions are more likely to have future relevance than idiosyncratic failures that might be one-off.

Accordingly, building on existing post-COVID analysis and reviews,⁴ we have concentrated our attention on monetary policy strategy. We focus on vulnerabilities in the policy framework that are especially relevant in the presence of large shocks, incomplete information, and rapidly changing risks. These relate to how decisions are made under uncertainty, how forecasts and scenarios are used, how risks are assessed and communicated, and how the framework can remain effective when outcomes differ sharply from the central projection. The resulting recommendations are intended to reinforce changes already made, support work underway, and improve preparedness for future shocks.

One aspect of the monetary policy strategy concerns the division of responsibility with fiscal policy when their objectives overlap, as they may do in crisis conditions. During COVID, these two arms of macroeconomic policy were indeed both aimed at supporting the economy and preventing a severe recession. We consider the interaction between the two policies.

We also assess the use of Additional Monetary Policy (AMP) instruments during the COVID period, and their general effectiveness (net of costs) at times when conventional policy instruments are unavailable or have become comparatively inefficient.

⁴ Including RBNZ (2022), McKibbin (2022), Schembri (2022), Chen and Richardson (2025), Conway (2025), Davies (2025).

1.2. Conclusions

Since its adoption in 1989, Inflation Targeting (IT) has served New Zealand well. Maintaining price stability is the most valuable contribution that monetary policy can make to economic progress and welfare. Implementation of IT evolved over time, becoming more flexible and less focused on price stability. The added flexibility contributed to vulnerabilities in policy strategy that can be exposed in challenging times such as the period under review.

The COVID epidemic threw up huge challenges. MPC's initial response was praiseworthy; rapid and, for New Zealand, innovative monetary accommodation was provided. The early adoption of a Least Regrets strategy helped get ahead of the potential for a compounding collapse and the quick rollout of Large Scale Asset Purchases (LSAPs) prevented market dysfunction from becoming part of the problem. Alongside an equally impressive fiscal response, Bank policy actions gave the New Zealand economy a timely and much-needed boost.

Though aware of the contingency for a long time, the Reserve Bank and the banks were insufficiently well prepared for policy near zero interest rates. Negative Interest Rate Policy (NIRP), a powerful and flexible tool, was inexplicably not ready for implementation. Some new policy tools were, later, rushed into action with insufficient forethought.

The unprecedented macroeconomic support and its unexpected effectiveness, together with resilience on the part of households and businesses, generated a stronger recovery than MPC was expecting. With ongoing uncertainty and concern about downside risks, it took a full year for policy to adjust to the reality that the initial economic stimulus had already achieved the intended result. Unexpectedly positive incoming data were discounted, and upside risks to the inflation outlook downplayed.

The uncertain environment also made it difficult to recognize that (more than) enough stimulus was in place and led to the Bank expanding the scale of LSAPs beyond what was needed. As a result, what began as a useful tool, later contributed to excess demand.

As inflation accelerated away from target, MPC decided to stop adding stimulus, but without sufficient urgency. The real policy interest rate – rarely discussed in MPC communications – continued to decline, and reached historic lows, even after the MPC communicated withdrawal of stimulus and hikes in the nominal Official Cash Rate (OCR).

As a result, the economy overheated to an historically extreme degree. Inflation rose as far as 7.3%, 5.3% above target, and the unemployment rate fell to 3.2%, an unsustainable historic low, necessitating a costly reduction in economic growth and employment to restore stability.

A core element of our message is that policy framework vulnerabilities that are modest in normally uncertain times can and did allow policy mistakes in times of great uncertainty. A different group of intelligent people, under the same framework, would risk making the same mistakes. Rather than data limitations or the absence of hindsight, we argue that vulnerabilities in policy strategy were the main factor.

Some contributing factors were unique to the period, but some vulnerabilities were present well before. The MPC inherited a flexible inflation targeting (FIT) strategy, implemented with Inflation Forecast Targeting (IFT), using a medium-term target horizon. This approach is vulnerable to policy errors under uncertainty. Policy errors are magnified when decisions are insufficiently focused on price stability and guardrails are not in place.

Indeed, ahead of COVID, Parliament and the Bank lowered several guardrails. The law was altered to introduce a Dual Mandate, adding scope for discretion and guiding the MPC to focus less on price stability. The Bank curtailed systematic scenario analysis, previously used to widen the decisionmaker's field of view. In the design of the new MPC's operating arrangements, an opportunity was missed to encourage diversity of thought – instead a preference for consensus decision-making and united external communication was chosen.

Measures to make the policy strategy more robust to uncertainty are available, as are numerous guardrails. Some have been adopted since the period under review. We offer recommendations for further consideration.

With regard to monetary and fiscal policy interactions, with the benefit of hindsight, the combined stimulus, while appropriate in the early stages, was too strong for too long. Early on, the fast evolution of fiscal policy and limits on projecting future fiscal policy challenged monetary policy formulation. The broader problem was not lack of information sharing or of integrated planning or coordination, but rather a misperception of how much stimulus was needed. The longstanding arrangement of dedicating each policy arm to different prime objectives and implementing policy separately remains the right one, though the Bank could incorporate its knowledge of the fiscal policy reaction function more actively into its own thinking.

The misperception of the amount of stimulus needed also complicates assessments of the value of AMP tools during COVID. Looking forward, however, some of these instruments should remain in the toolkit; new variants should be considered; and regular work should be done to ensure their readiness for implementation.

An additional vulnerability, unrelated to the IFT strategy and not active during the period under review, deserves highlighting. In the Reserve Bank of New Zealand Act 2021 are provisions that could place a financial constraint on monetary policy's ability to pull the economy out of recession. The concern for fiscal soundness is reasonable, but the specific design is problematic.

1.3. Recommendations

The review identified opportunities to strengthen the ongoing effectiveness of monetary policy in New Zealand. The recommendations included here build on an earlier review by the RBNZ covering most of the same period, and comments by Warwick McKibbin and Lawrence Schembri.

1. The MPC should develop a **monetary policy strategy that addresses known vulnerabilities and is more robust to uncertainty** than in the period under review.
2. **Systematic policy:** The MPC should formulate monetary policy in a more systematic manner and rely less on discretion than observed in the period under review. Constraining discretion is key to successful Inflation Targeting.

3. **Greater emphasis should be placed on real interest rates** in assessing and communicating the stance of monetary policy. The Monetary Policy Statement (MPS) should include real OCR paths implied by the nominal OCR paths and the associated inflation projections.
4. **Near-term inflation signals.** Near-term inflation developments should be more in focus. Near-term indicators may be more robust than forecasts at the policy horizon, which are often uninformative.
5. The **policy rule in the forecasting model** is an integral part of MPC strategy. MPC should agree on the specification of this rule for reasons of consistency in the preparation of forecasts and greater transparency on discretionary adjustments to policy.
6. Prescriptions from one or two **simple rules** selected for their robustness characteristics in the New Zealand context should be presented in the MPS for the current and next quarter. This serves as a crosscheck on current policy.
7. **Scenario analysis:** Renewed focus on scenarios is welcome. Scenarios should be sufficiently broad to examine key assumptions and judgements, and to cover upside and downside risks to the outlook.
8. **Operations at the Effective Lower Bound (ELB):** AMP instruments should be periodically reviewed and tested to ensure continuing operational readiness for implementation in an evolving financial landscape.
9. **Fiscal policy:** When forecasting or constructing scenarios, endogenous fiscal responses to significant economic developments should be allowed for.
10. Modifications to the MPC Charter to encourage voting and public discussion of individual views are welcome. MPC should **institutionalise evaluation of a broader distribution of plausible policy paths**.
11. **Staff resources.** The Bank should allocate sufficient resources to support the effective functioning of the MPC, including policy research and analysis to support diverse perspectives in policy deliberations.
12. **Sections 121 and 208 of the RBNZ Act** create risks to monetary stability that should be reevaluated. The RBNZ Board should not have the authority to refuse to implement the MPC's monetary policy.

2. The evolution of Inflation Targeting

Monetary policy, executed well, can provide the foundation of monetary stability that is necessary for economic progress, and a cushion during periods of crisis. Executed poorly, it can become a source of instability, undermining growth and prosperity. Over the decades, New Zealand has experienced both. It pioneered Inflation Targeting (IT) after poor monetary performance, during much of the 1970s and 1980s.⁵ Attempts at monetary restraint failed repeatedly. Though price stability was one goal, inflation was high and volatile. Multiple, often conflicting, objectives overburdened monetary policy.

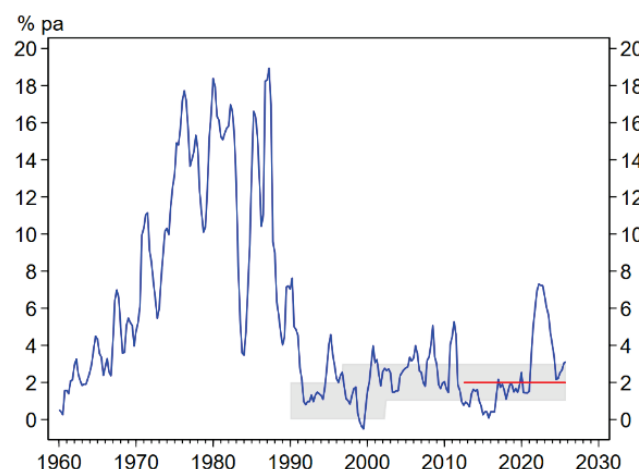
⁵ Additional information regarding the evolution of monetary policy in New Zealand, including the adoption and evolution of Inflation Targeting, can be found in Nicholl and Archer (1992), Reddell (1999), Archer (2000), Bollard and Karagedikli (2006), Graham and Smith (2012), McDermott and Williams (2018) and Buckle (2018).

The Reserve Bank of New Zealand Act 1989 clarified objectives and fundamentally altered the institutional structure for monetary policy decisions. Section 8 of the Act identified the achievement and maintenance of price stability as the Bank’s primary monetary-policy objective.

“The primary function of the Bank is to formulate and implement monetary policy directed to the economic objective of achieving and maintaining stability in the general level of prices.” (New Zealand Parliament, 1989.)

The Bank’s Governor was granted operational independence with the authority and the responsibility to pursue price stability. The price stability goal was given tangible form with a quantitative inflation target. This institutional arrangement worked. Since the early 1990s, inflation has generally been low and stable (Figure 2.1) and indicators of real economic activity suggest greater economic stability compared to the earlier period (Figure 2.2).

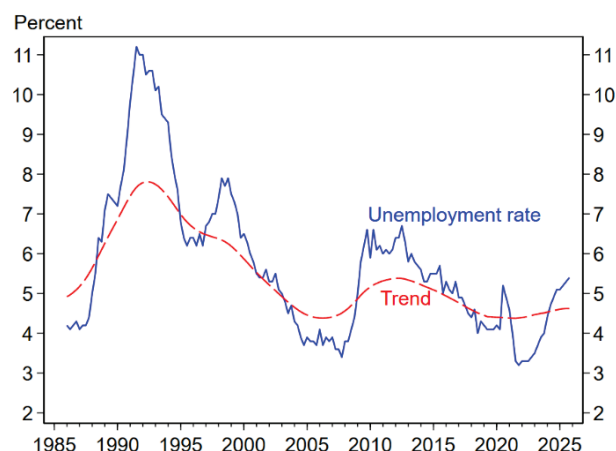
Figure 2.1 – Annual CPI inflation



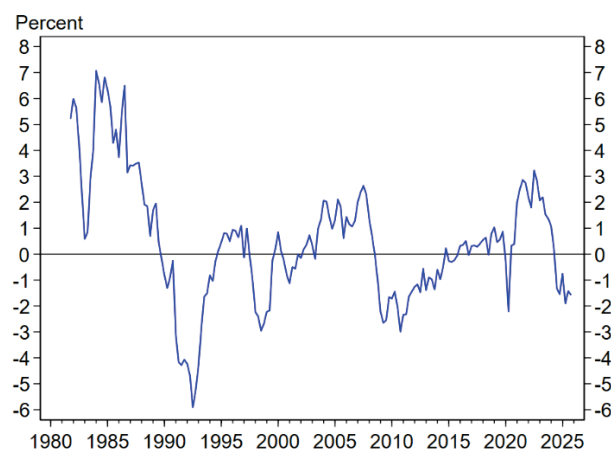
Notes: Shaded area indicates target range since the adoption of IT. Red line indicates focus on midpoint as of 2012q3.

Figure 2.2 – Measures of real economic activity

Unemployment rate and trend estimate



Output gap



Notes: Quarterly unemployment rate (seasonally adjusted) and RBNZ estimate of trend/normal unemployment rate. Output gap is the percent difference of actual GDP from RBNZ current estimate of potential GDP.

Operationally, IT can be implemented in various ways, varying in simplicity and analytical requirements. The approach developed by the Bank used inflation forecasts effectively as intermediate targets—IT was Inflation *Forecast* Targeting (IFT). As forecasts account for policy transmission lags, policy planned this way mainly responds to inflation deviations from the inflation target that are expected to persist in the absence of counteracting policy action. By not responding to transitory inflation shocks, unnecessary volatility in economic activity can be avoided. The primacy of price stability did not imply indifference to developments in the real economy—no IT central bank behaved as an “inflation nutter” (King, 1997).

Formal modelling of IT appeared to capture the IFT implementation.⁶ But the modelling approach used also suggested that tolerating greater instability in inflation would promise greater stability in output. These more “flexible” strategies became known as Flexible Inflation Targeting (FIT, Svensson, 1999a,b).⁷ Greater flexibility meant that the central bank could aim to bring inflation back to target more gradually after a shock. In theory, optimal policy under FIT implied a choice on the so-called efficiency frontier, as illustrated in Figure 2.3, reproduced from the second version of the Reserve Bank’s Monetary Policy Handbook (RBNZ, 2020).

Figure 2.3 – The volatility tradeoff



Source: Reproduced from RBNZ Monetary Policy Handbook, version 2, September 2020. Notes: The public indifference curve plots all the combinations of output and inflation volatility that the public (or the central bank representing the public) are indifferent between. The monetary policy efficiency frontier plots all of the combinations that can feasibly be achieved with monetary policy.

There is, however, an important qualification. Greater flexibility in theoretical models gives nice results when: everyone understands the economy’s structure, its potential output, and the monetary policy transmission mechanism, all with considerable precision; expectations are rational; and the public fully understands monetary policy’s objectives. With such perfect knowledge the central bank can shift towards stabilizing an estimated output gap without endangering the nominal anchor even as inflation forecast deviations from target are tolerated over longer horizons. In reality, knowledge is imperfect, and beliefs of households and firms may reflect approximations and rules of thumb to varying degrees. Moreover, beliefs need to be updated as new data reveal perception errors, with the learning process also a source of wedges between the textbook world and reality.⁸

Such perception errors and imperfect updating can substantially change the shape and positioning of the efficiency frontier described above. Figure 2.4 illustrates the conundrum. Volatility curves are plotted under the alternative assumptions of perfect knowledge and of imperfect knowledge. The former is similar to the monetary policy efficiency frontier plotted in Figure 2.3. Points T and F mark the points on the respective frontiers corresponding to more (F) and less (T) flexible implementations of inflation targeting. Under perfect knowledge, both points appear reasonable, with F preferred if greater output stability is desired. Given imperfect knowledge, however, points F and T are not feasible. Volatility is worse on both dimensions, the more so when policy is implemented flexibly. Notably, an attempt to achieve F (based on an incorrect assumption of perfect knowledge) instead destabilizes both inflation and output, as shown with point F*. Less flexible implementations result in lower volatility in *both* the inflation and output dimensions, shown with point T* in the figure. The reason is that under imperfect knowledge inflation expectations are never perfectly anchored, and greater deviations from target compromise credibility.

⁶ Svensson (1997) presents a classic treatment. In 2001, Svensson completed an independent review of monetary policy in New Zealand (Svensson, 2001).

⁷ Since this modelling policy approach suggests closing both the inflation gap and output gap, McCallum and Nelson (1998) observed that “flexible inflation targeting” could also be described as “flexible output-gap targeting” (p. 35).

⁸ Section 5 further discusses “imperfections” in the economy and policy economists’ understanding of it, in the context of the RBNZ’s policy forecasts made during the period under review.

The essential point is that good intentions can produce bad outcomes when the underlying conditions are not as benign as presumed. The desire to limit unemployment variations around trend level can, by slowing reactions to emerging inflation, *increase* the variability of unemployment. Flexibility that appears desirable under ideal modelling assumptions can be excessive and prove counterproductive in practice.⁹

Over time the Reserve Bank described its IT framework with terms suggesting greater flexibility. In part, this reflected the Reserve Bank's improved credibility. Credibility allows greater flexibility even as price stability remains the primary goal. But the 2018 amendment of the Reserve Bank Act took a significant further step towards flexibility not based on additional credibility. It changed the Bank's objectives and the institutional arrangements that had prevailed since the 1989 Act. Price stability would no longer serve as the Bank's primary goal. Instead, the Bank would operate with a dual mandate, as described in Section 8(1) of the revised Act:

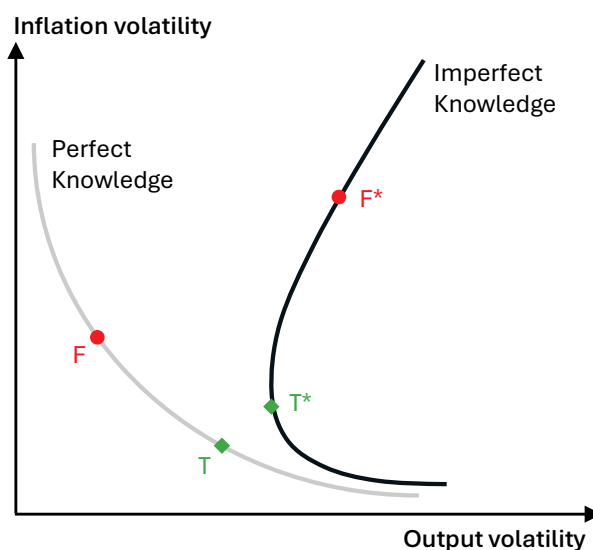
“The Bank, acting through the MPC, has the function of formulating a monetary policy directed to the economic objectives of—

- (a) achieving and maintaining stability in the general level of prices over the medium term; and
- (b) supporting maximum sustainable employment.” (New Zealand Parliament, 2018).

These amendments downplayed the importance of price stability, placed a larger weight on output and employment stability, so effectively mandated that the Bank should tolerate larger and longer lasting inflation deviations from target. In August 2019, just a few months after it started operating, the new MPC described the MPC's “flexible inflation targeting framework” in terms consistent with an increase in flexibility (Box A, Monetary Policy Statement, August 2019). This strategy statement was in effect at the start of the period under review and remained in effect, with some adaptation, throughout.

As for institutional arrangements, the 2018 Act shifted responsibility for monetary policy decisions from the Governor to the newly created MPC and moved target specification (the Remit) from being subject to the Governor's agreement to a unilateral Ministerial directive.

Figure 2.4 – The volatility tradeoff under perfect and imperfect knowledge



Notes: Volatility tradeoff expected under the assumption of perfect knowledge and outcomes when expectations are formed under imperfect knowledge. Illustration based on Orphanides and Williams (2004).

⁹ The perceived short-run tension between price stability and economic stability has been evident in New Zealand from the very beginning of Inflation Targeting. The RBNZ explained that focusing on price stability was ultimately about promoting New Zealand's economic and social welfare: “The Act is based squarely on the weight of New Zealand and international evidence which suggests that price stability is the single most valuable contribution that monetary policy can make to New Zealand's economic and social welfare, and the only contribution which is likely to endure.” (RBNZ, Monetary Policy Statement, September 1990).

3. What happened?

To evaluate monetary policy in the period under review, it is instructive to examine economic indicators and policy decisions from two perspectives: a real-time perspective, accounting for the information and analysis available to policymakers when decisions were made; and in retrospect, based on best available data and information that has subsequently become available. Given the lags associated with monetary policy, evaluation of policy in the period under review can also be informed by examining policy over a somewhat broader window. In this section we provide a brief and selective summary for the period under review and surrounding years.¹⁰

Figure 3.1 presents a retrospective summary of economic developments and interest rates from 2018Q1 to 2025Q4, with vertical lines at 2020Q1 and 2022Q4 highlighting the period under review. Annex 1 provides a timeline of events, major economic news, and RBNZ decisions.

When the MPC started operating, on 1 April 2019, inflation was slightly below the 2% target and GDP was somewhat above estimates of its potential. According to current RBNZ estimates, the output gap was positive during the second half of 2018 and 2019, averaging just below 1%.¹¹ MPC eased policy at its very first policy meeting, on 8 May 2019, lowering the OCR by 25 basis points to 1.5 percent, then by 50 basis points, on 7 August 2019.

On 11 March 2020, the World Health Organization declared a pandemic, prompting a series of decisions by authorities around the world, responding to the health crisis and protecting the economy in their respective jurisdictions. Fiscal and monetary policy were eased massively in most advanced economies, including in New Zealand. The MPC cut the OCR further, to 0.25%, and implemented additional monetary policy easing measures that are discussed later. The RBNZ also used its powers as a prudential authority to ease regulatory measures, thereby providing additional support to aggregate demand.

The pandemic shock resulted in a sharp upward spike in the unemployment rate (and downward spike in the output gap) and disinflationary pressure. In 2020Q2, the CPI fell by 1.3 percent at a seasonally adjusted annual rate (saar). While the health emergency was not effectively resolved until the end of 2021, the economy recovered remarkably quickly. Based on the RBNZ's current estimates, the output gap turned positive as early as 2020Q3 and the unemployment rate fell below its sustainable level by 2021Q2. By 2020Q4, quarterly inflation (saar) had already risen above the inflation target; it accelerated in subsequent quarters.¹² Meanwhile, the OCR remained at 25 basis points until the second half of 2021, and well below its estimate of the neutral nominal interest rate until 2022. LSAPs and other AMP tools added further stimulus. With inflation rising rapidly, the MPC effectively kept reducing the real interest rate even as targets were being breached, leading to severe overheating of the economy.

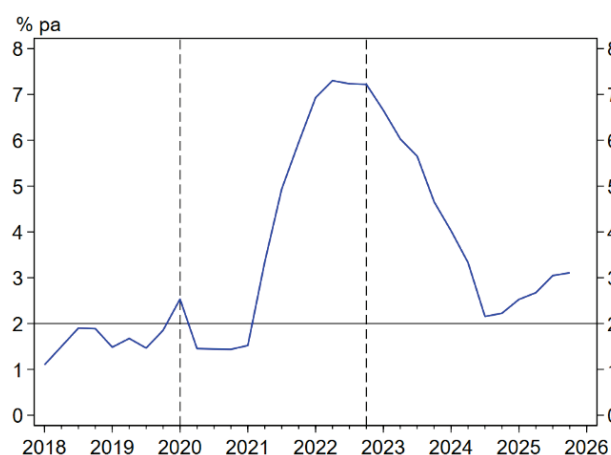
¹⁰ RBNZ (2022) provides the Reserve Bank's own review of monetary policy in the 2017-2022 period, including a detailed description of policy actions in that period. A more recent perspective, including a description of later developments, appeared in Conway (2025) and Davies (2025).

¹¹ References to "current" or "revised" information refer to data and estimates available with the publication of the May 2026 MPS.

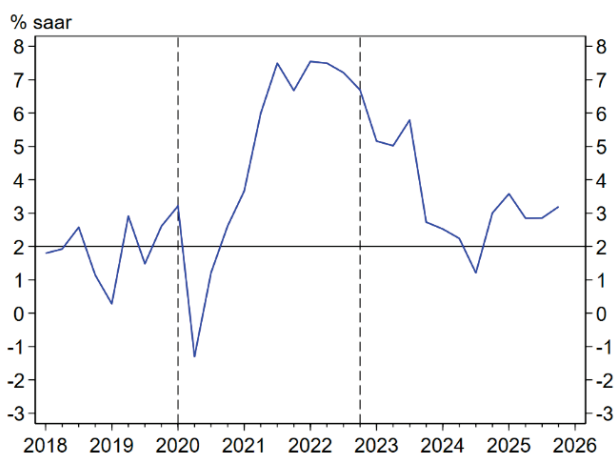
¹² This is easier to see in the quarterly series than in the annual inflation series that is preferred by the RBNZ for communication and analysis. The annual series represents a 4-quarter moving average. The short-lived deflation observed in 2020Q2 obscured the acceleration in the annual series until that quarter dropped out of the moving average a year later.

Figure 3.1 – What happened?

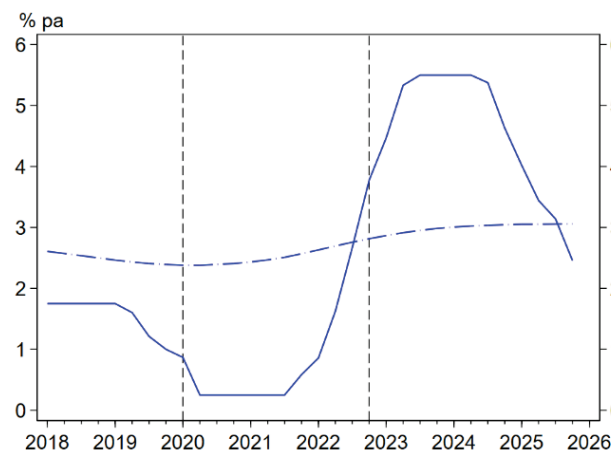
Annual inflation



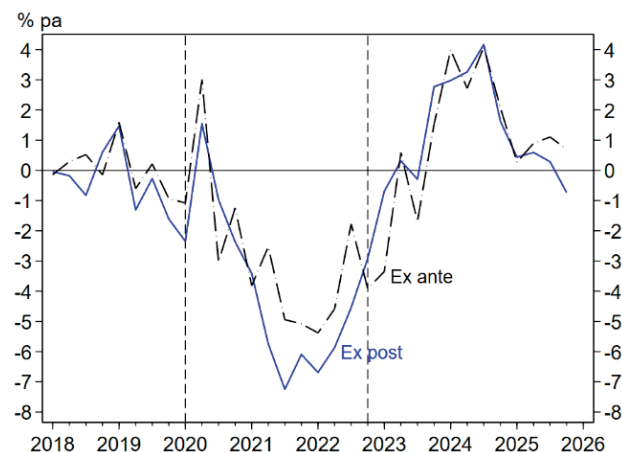
Quarterly inflation (SAAR)



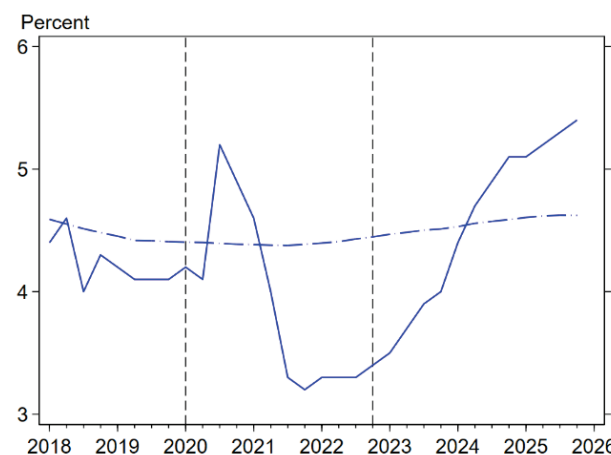
Nominal OCR



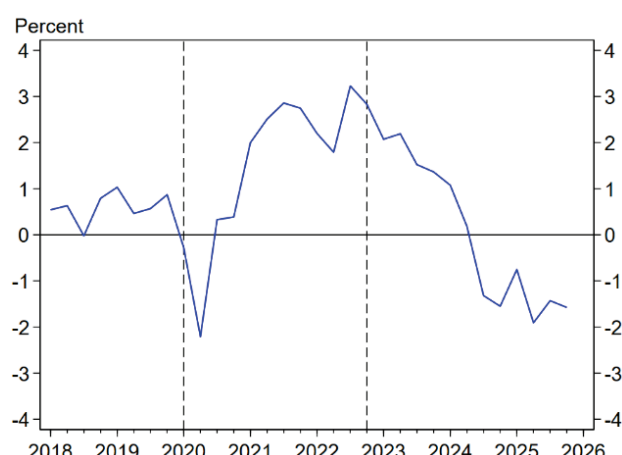
Real OCR



Unemployment rate



Output gap



Notes: Blue lines reflect outcomes and the RBNZ's trend estimates as of May 2026. Ex post real rate defined as OCR minus quarterly inflation. Ex ante real rate defined as OCR minus the RBNZ's nowcast of quarterly inflation. Vertical lines at 2020q1 and 2022q4.

The unemployment rate reached a cyclical low of just 3.2 percent in 2021Q4 while the output gap reached its peak at just over 3 percent in 2022Q3. Quarterly inflation reached 6 percent (saar) in 2021Q2 and stayed above that level until 2023.¹³

MPC eventually responded to the accelerating inflation by first halting LSAPs then increasing the OCR, initially cautiously in late 2021 then aggressively. In 2023, the real OCR also rose substantially, thereby tightening monetary policy sufficiently to slow the economy and eventually bring inflation back under control. The economic cost of regaining control of price stability was high. The unemployment rate rose sharply during 2023 and 2024 and in 2025 surpassed the 5.2 percent peak it had reached during the pandemic.

This retrospective summary suggests that monetary policy was not appropriately calibrated during the period under review. It suggests that the monetary policy accommodation provided was excessive, and that this policy error was corrected too late, leading to a costly boom-bust cycle.¹⁴

While the evolution of the economy and interest rate policy presented in Figure 3.1 suggests a policy error, assessing whether MPC decisions were problematic requires accounting for the information available to the MPC. To evaluate whether MPC decisions were reasonable, it is essential to focus on the information and analysis available to policymakers in real time, when decisions were made. To that end, it is helpful to examine the evolution of RBNZ projections and analysis available to the MPC at the time of the policy meetings.

Figure 3.2 presents a graphical summary of the evolution of baseline projections in this period. In each panel, the blue solid lines present economic outcomes for the variables shown except for the ex ante real OCR. To reflect more closely the real OCR corresponding to MPC decisions at MPS meetings, the ex ante real OCR shown in the blue line is defined as the nominal OCR minus the nowcast of quarterly inflation (saar) corresponding to that MPS.¹⁵ The remaining lines, in each panel, present quarterly vintages of forecasts published in the MPS from February 2020 to November 2022. For the ex ante real OCR, each vintage shows the difference between the OCR path and the quarterly inflation path (saar) corresponding to each MPS.

The February 2020 forecast round was completed before the pandemic shock materialized. As can be seen, the economy was assessed to be in a goldilocks scenario. The Summary Record of Meeting noted that “the coronavirus situation was a complicating factor given how quickly it was changing and the limited information available.” (February 2020 MPS.)

The May 2020 forecast round was the first to reflect the economic consequences of the pandemic shock. By then, the MPC had already taken significant policy easing action. The baseline projection was quite pessimistic. The unemployment rate was projected to rise to 7 percent in 2020Q2 and to 9 percent in 2020Q3. Risks were assessed to the downside, and two additional, more pessimistic, scenarios were also discussed.

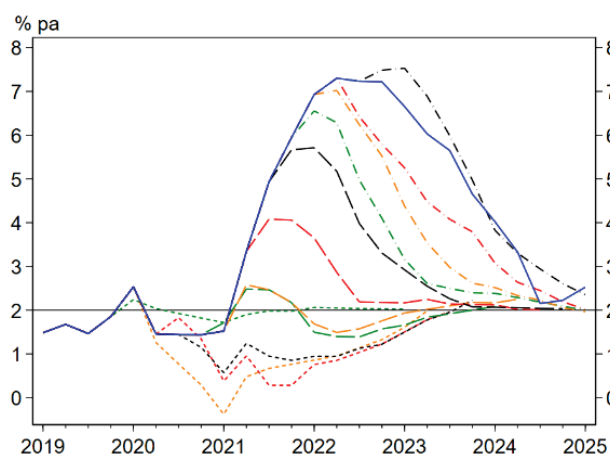
¹³ Annual inflation peaked at 7.3 percent in 2022Q2. In addition to the severe overheating of the economy through 2021, this reflected a sharp rise of energy prices when war started in Ukraine in 2022Q1. A similar profile is shown by a range of underlying and core inflation measures.

¹⁴ The RBNZ has acknowledged that the policy stimulus was removed too late. (RBNZ, 2022, Conway, 2025.)

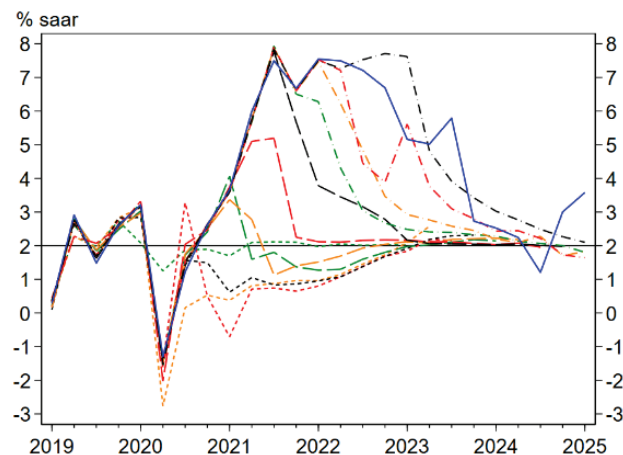
¹⁵ The resulting real OCR is an approximation since the nominal OCR represents the quarterly average of daily rates and not a three-month rate at the start of the month.

Figure 3.2 – Evolution of RBNZ projections: February 2020–November 2022

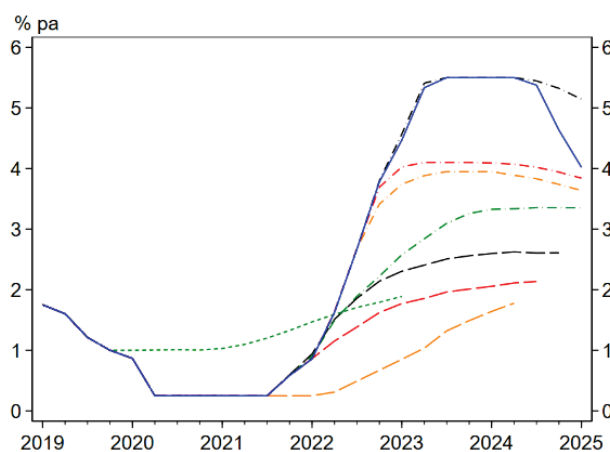
Annual inflation



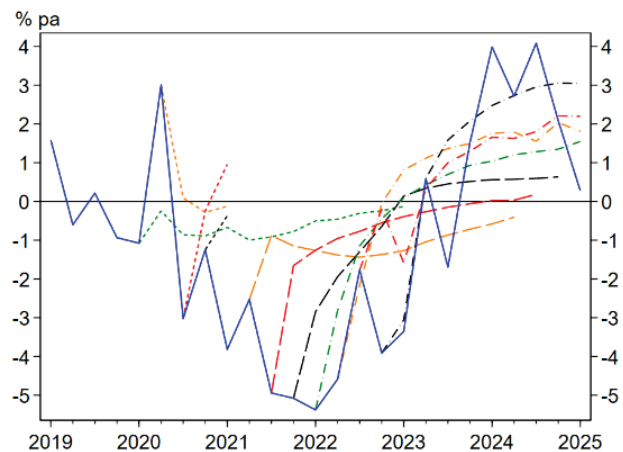
Quarterly inflation (SAAR)



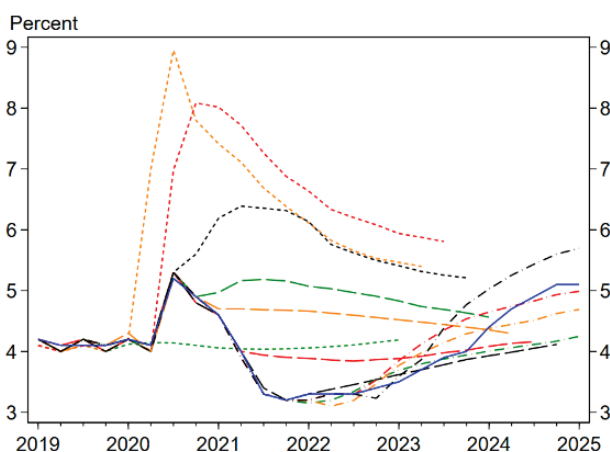
Nominal OCR



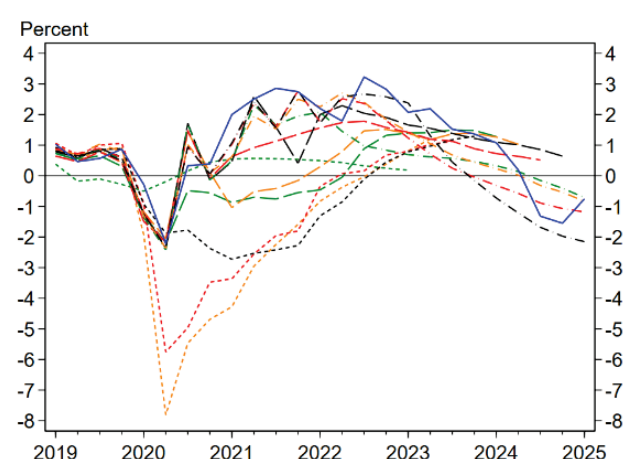
Real OCR



Unemployment rate



Output gap



Notes: Green, orange, red, and black lines correspond to Feb., May, Aug., and Nov. MPS data and projections. Short dashes, long dashes, and dash-dots correspond to 2020, 2021 and 2022 vintages, respectively. Solid blue lines reflect outcomes as of May 2026 except for the ex ante real OCR that is defined as the nominal OCR minus the nowcast of quarterly inflation.

With regard to inflation, the nowcast for 2020Q2 suggested a 2.7 percent decline in quarterly CPI (saar). Low positive inflation, below 1 percent (saar), was projected until 2022Q2. The MPC communicated that “considerable monetary stimulus remains necessary to achieve our objectives.” (May 2020 MPS, p. 10.) In addition, the Summary Record of Meeting provided information about MPC’s “least regrets” policy approach that shaped policy for several quarters:

“Members agreed that a ‘least regrets’ monetary policy approach is needed, delivering stimulus sooner rather than later, and thus minimising the risk that the stimulus delivered turns out not to be enough.” (p. 5, May 2020 MPS.)

Faced with such dire economic projections, and having adopted a policy approach favouring even more stimulus for risk management considerations, the monetary stimulus provided early in the pandemic was reasonable.

In the event, incoming data suggested that the economy was more resilient than had been projected, and the stimulus was more effective. This is reflected in the evolution of projections for the unemployment rate and the output gap in the figure. The projected peak in the unemployment rate was revised from 9 percent in the May 2020 MPS to 8.1 percent in the August 2020 MPS and 6.4 percent in November 2020 MPS. Further improvements in economic prospects persisted through the 2021 forecast rounds. In line with the revisions to the outlook for economic activity, near-term inflation projections were uniformly adjusted upwards. Inflation projections at more distant horizons invariably converged to the 2 percent target, effectively by construction.¹⁶

Overall, starting with the May 2020 forecast round, the figure suggests that this episode is characterized by large and persistent forecast errors. The available forecasts were generally considerably more pessimistic than realised outcomes and encouraged a far greater degree of monetary stimulus than turned out to be needed.

To some extent, the policy error identified earlier can be attributed to flawed forecasts. The forecasts in this episode suggest the persistence of excessive pessimism that resulted in large serially correlated errors in both inflation and economic activity. While not unusual in small samples, the size and serial correlation of forecast errors in the few quarters after the pandemic shock stand out.

Forecast errors are unavoidable, especially in episodes characterized by such extreme uncertainty as in 2020.¹⁷ That said, the magnitude of earlier forecast errors was revealed with subsequent releases of macroeconomic data. This information would be expected to lead to a correction in subsequent forecasts and, importantly, inform the MPC about the direction of policy adjustment suggested by the data releases relative to the assessment of policy before the forecast errors were revealed. Persistent positive surprises to inflation and real economic activity would ordinarily suggest that a reduction in stimulus is warranted.

¹⁶ Inflation forecasts towards the end of the forecast horizon can converge to a specified target regardless of the policy decision at the policy meeting corresponding to the forecast round. This can be done as part of a strategy discussion or automatically. At RBNZ convergence is automatically incorporated in the forecasting process, using an algorithm referred to as the “forecasting rule.” The algorithm introduces unanticipated shocks to the OCR in future quarters so that the inflation forecast towards the end of the forecast horizon converges to the specified target.

¹⁷ See Knowles and Patel (2025) for a recent assessment of RBNZ forecast errors. Other central banks were similarly challenged in this episode, see e.g. Bernanke (2024) and Orphanides (2023).

Table 3.1 lists the quarterly macroeconomic announcements for CPI, GDP and the unemployment rate in the period under review. For each announcement, it records the outturn and associated surprise when compared to the latest projection published by the RBNZ before the data release. The last column notes the impact the surprise would normally have been expected to have on interest rate policy. Surprises from the beginning of the pandemic until 2022Q2 were mostly positive, which typically would have been expected to point towards tighter policy. Was this reflected in revisions to the interest rate policy?

Table 3.1 – Macroeconomic announcements

	Release date	Data	Outturn	Last forecast	Surprise	Effect on OCR path
2020	24-Jan-20	Q4 CPI	1.9	1.6	0.3	↑
	5-Feb-20	Q4 Unemployment	4.0	4.2	-0.2	↑
	19-Mar-20	Q4 GDP	2.3	2.2	0.1	↑
	20-Apr-20	Q1 CPI	2.5	2.2	0.3	↑
	6-May-20	Q1 Unemployment	4.2	4.1	0.1	↓
	18-Jun-20	Q1 GDP	1.5	1.3	0.3	↑
	16-Jul-20	Q2 CPI	1.5	1.3	0.2	↑
	5-Aug-20	Q2 Unemployment	4.0	7.0	-3.0	↑
	17-Sep-20	Q2 GDP	-2.1	-2.7	0.5	↑
	23-Oct-20	Q3 CPI	1.4	1.8	-0.4	↓
	4-Nov-20	Q3 Unemployment	5.3	7.0	-1.7	↑
	17-Dec-20	Q3 GDP	-2.3	-3.0	0.7	↑
2021	22-Jan-21	Q4 CPI	1.4	1.1	0.3	↑
	3-Feb-21	Q4 Unemployment	4.9	5.6	-0.7	↑
	18-Mar-21	Q4 GDP	-3.0	-2.7	-0.3	↓
	21-Apr-21	Q1 CPI	1.5	1.7	-0.2	↓
	5-May-21	Q1 Unemployment	4.7	5.0	-0.3	↑
	17-Jun-21	Q1 GDP	-2.3	-3.1	0.8	↑
	16-Jul-21	Q2 CPI	3.3	2.6	0.8	↑
	4-Aug-21	Q2 Unemployment	4.0	4.7	-0.7	↑
	16-Sep-21	Q2 GDP	5.1	4.0	1.1	↑
	18-Oct-21	Q3 CPI	4.9	4.1	0.9	↑
	3-Nov-21	Q3 Unemployment	3.4	3.9	-0.5	↑
	1-Dec-21	Q3 GDP	5.0	3.7	1.3	↑
2022	27-Jan-22	Q4 CPI	5.9	5.7	0.3	↑
	2-Feb-22	Q4 Unemployment	3.2	3.2	0.0	→
	17-Mar-22	Q4 GDP	5.6	5.5	0.2	↑
	21-Apr-22	Q1 CPI	6.9	6.6	0.4	↑
	4-May-22	Q1 Unemployment	3.2	3.2	0.1	↓
	16-Jun-22	Q1 GDP	5.1	5.4	-0.3	↓
	18-Jul-22	Q2 CPI	7.3	7.0	0.3	↑
	3-Aug-22	Q2 Unemployment	3.3	3.1	0.2	↓
	15-Sep-22	Q2 GDP	1.0	1.2	-0.2	↓
	18-Oct-22	Q3 CPI	7.2	6.4	0.8	↑
	2-Nov-22	Q3 Unemployment	3.3	3.3	0.0	↓
	15-Dec-22	Q3 GDP	2.7	2.4	0.2	↑

Source: RBNZ historical forecasts. Notes: Last forecast shows the latest RBNZ MPS projection prior to the release of the data shown. GDP is annual average percent change; CPI inflation is annual % change; and unemployment is % of the labour force. GDP growth outturns shown may be a rounding error out from StatsNZ numbers.

The panel in Figure 3.2 comparing projections of the nominal OCR shows that some upward revision to the OCR path occurred from 2021Q2 onwards.¹⁸ However, the panel showing the ex ante real OCR path suggests that these revisions to the nominal OCR path were too small. In 2021Q2, the whole path for the real OCR was negative (yellow dashed line). In 2021Q3, 2021Q4 and 2022Q1, the MPC decisions to cautiously increase the OCR corresponded to further reductions of the current-quarter real OCR.

While the accommodation provided early in the pandemic appears reasonable on the basis of real-time projections and risks, the subsequent series of positive news and evolution of projections called for a more timely policy normalization and reversal than was actually observed.

4. Were better economic outcomes feasible?

The data presented above suggests that monetary policy was excessively accommodative during the recovery from the pandemic, leading to an overheating of the economy in 2021 and 2022. This contributed to a spike in inflation that subsequently necessitated a substantial tightening of monetary policy that raised the unemployment rate and curtailed growth in order to restore price stability.

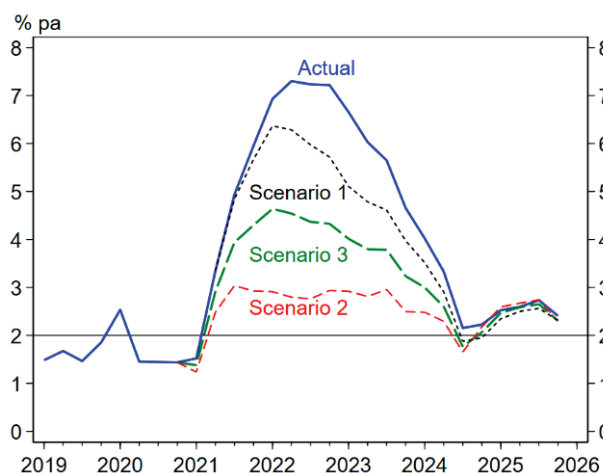
Could different policy decisions have resulted in better economic outcomes—a lower spike in inflation and greater economic stability with a less pronounced boom-bust cycle? Existing counterfactual analysis by the Bank can be used to help answer this question. Cherry and Kendall (2025) present two counterfactual monetary policy scenarios using NZSIM, the model that was being used to guide the forecasting process and policy analysis in the period under review. Scenario 1 examines the path the economy would have taken if policy normalization had taken place somewhat earlier than it actually did. Scenario 2 examines the economy's counterfactual path if policy tightening had started much earlier, and furthermore, OCR hikes had been sufficiently forceful to contain the rise of inflation to 3 percent over the period. Figure 4.1 reproduces these two scenarios from their work. It also adds a third: Scenario 3, constructed as an intermediate policy option to scenarios 1 and 2.¹⁹ The OCR paths corresponding to the three scenarios, as well as the actual OCR path, are shown in the panel depicting the nominal OCR in the figure. Remaining panels in the figure compare the respective evolutions of inflation, economic activity and the real OCR to what actually happened.

¹⁸ This corresponds to the May 2021 MPS. In earlier quarters, from the May 2020 MPS to the February 2021 MPS, the OCR path was truncated in 2021Q1, and only reflected the forward guidance provided in March 2020 that the OCR would remain at 25 basis points for one year.

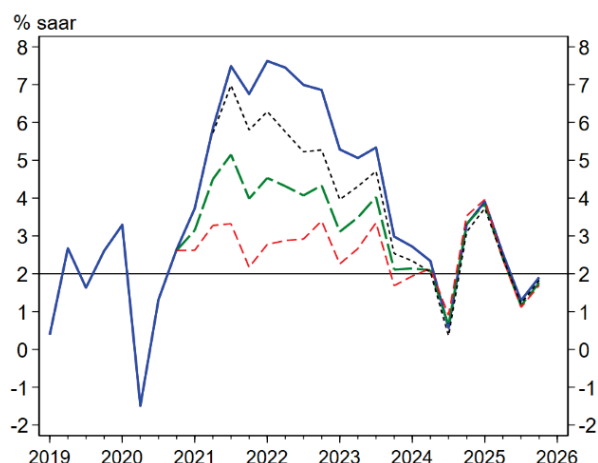
¹⁹ Intermediate scenarios can be constructed using the results of scenarios 1 and 2 reported in Cherry and Kendall (2025) due to the linearity properties of the underlying model employed for the simulations.

Figure 4.1 – Simulating alternative policy options

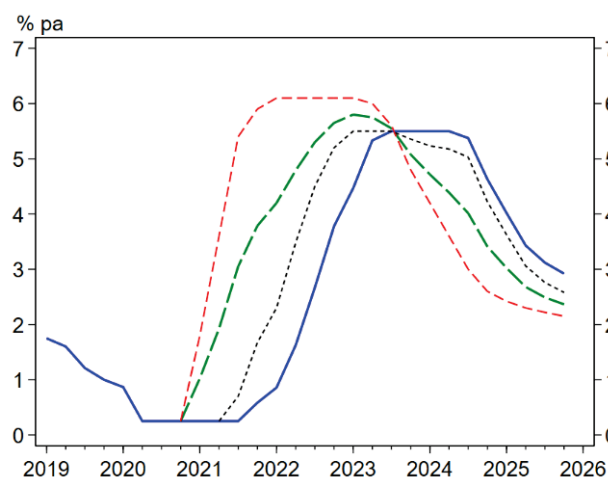
Annual inflation



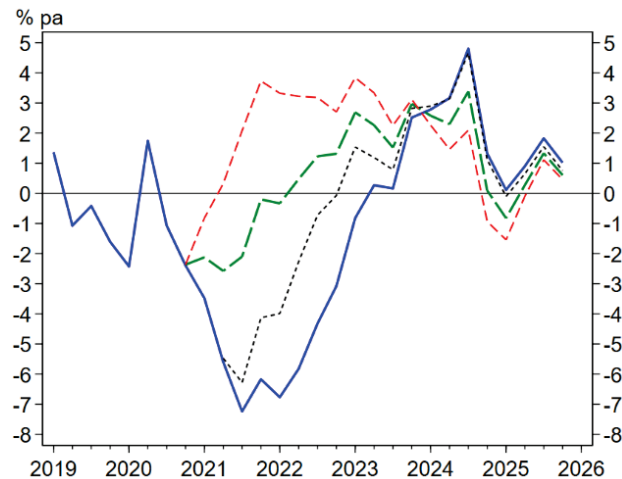
Quarterly inflation (SAAR)



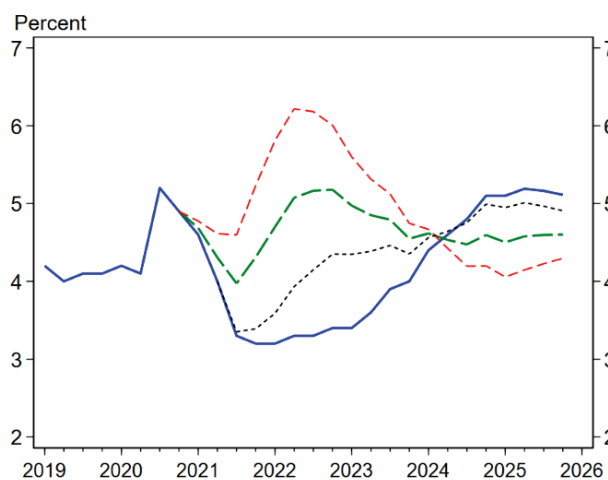
Nominal OCR



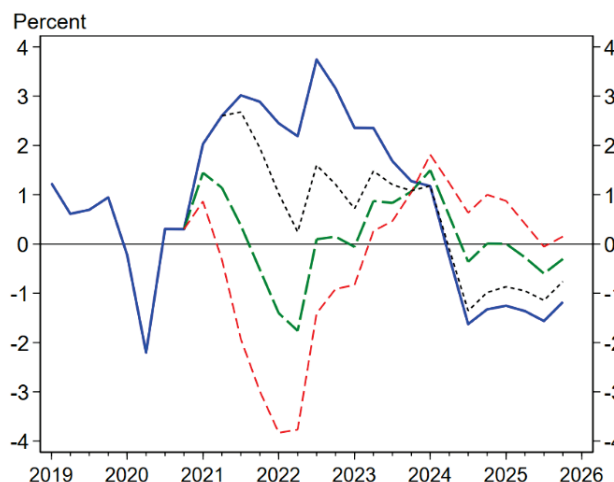
Real OCR



Unemployment rate



Output gap



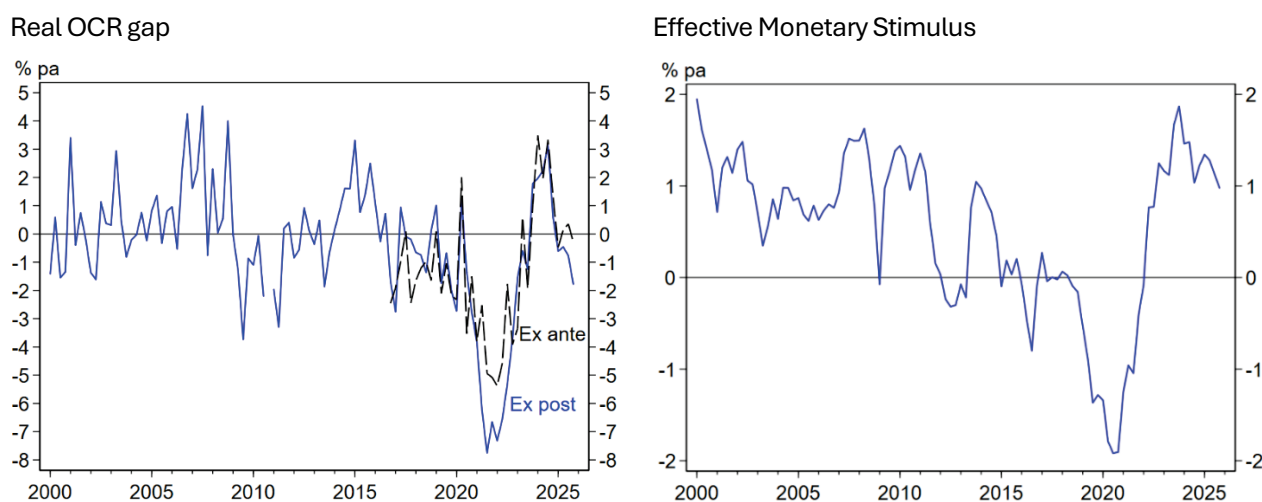
Notes: Alternative policy options based on RBNZ model simulations. Actual, scenario 1 and scenario 2 as described in Cherry and Kendall (2025). Scenario 3 is the intermediate policy option to scenarios 1 and 2. Real OCR is defined as nominal OCR minus quarterly inflation.

The alternative scenarios are quite informative. Scenario 1 confirms that an earlier but similarly-paced policy normalization would have produced somewhat better economic outcomes, but not dramatically so. Annual inflation would still have risen substantially, above 6 percent. The boom-bust cycle that resulted in rising unemployment after the pandemic would still have materialized, though somewhat muted. Scenario 2 is also instructive, for a different reason. This counterfactual confirms the role of adverse supply shocks in explaining part of the rise in inflation during this episode. The policy tightening required to contain inflation below 3 percent in the presence of these shocks would have required a severe economic downturn. This alternative policy option would not have resulted in better economic outcomes, compared to actual policy.

In contrast to scenarios 1 and 2, the counterfactual simulation of scenario 3 shown in the figure suggests that if policy normalization had started earlier, but progressed less aggressively than in scenario 2, significantly better economic outcomes would have been observed. Annual inflation would have peaked below 5 percent, but most importantly, the boom-bust cycle would have been avoided. The real OCR path reveals the key difference. In contrast to actual policy and scenario 1, policy under scenario 3 would have adjusted the nominal OCR so as to prevent the real OCR from falling and the economy from overheating during 2021, but not so much as to cause a severe downturn, as in scenario 2. Policy under scenario 3 would have contained the rise in inflation and also kept the unemployment rate closer to its maximum sustainable level. Had the MPC implemented policy as in scenario 3, it would have fulfilled its dual mandate considerably better.

Figure 4.2 puts the stimulus provided in the post-pandemic episode in historical context. The left panel shows two measures of the real OCR gap. The ex-ante measure adjusts the nominal OCR by subtracting the nowcast of quarterly inflation (saar) available to the MPC in each MPS quarter, and the real-time estimate of the equilibrium real interest rate. The ex post measure uses the actual inflation rate during the quarter (saar) and the ex post revised estimates of the equilibrium real interest rate. By either measure, interest rate policy was extraordinarily accommodative around the end of 2021.

Figure 4.2 – Measures of monetary policy: 2000-2025



Notes: Left panel plots real OCR gaps excluding the observation for 2010Q4 that is distorted due to the GST increase in that quarter. Ex ante estimate based on nowcast of inflation and real-time estimate of r^* . Ex post estimate based on actual inflation and retrospective estimates of r^* as of May 2026. Right panel plots RBNZ estimates of the effective monetary stimulus.

The right panel presents the RBNZ's recent estimates of the Effective Monetary Stimulus (EMS). EMS is an alternative measure of monetary policy stimulus that summarizes the stance of monetary policy across the yield curve, and more consistently over time when the sample includes a period when the policy rate is constrained at its lower bound.²⁰ This measure reinforces that the RBNZ provided an extraordinary degree of accommodation in this episode.²¹

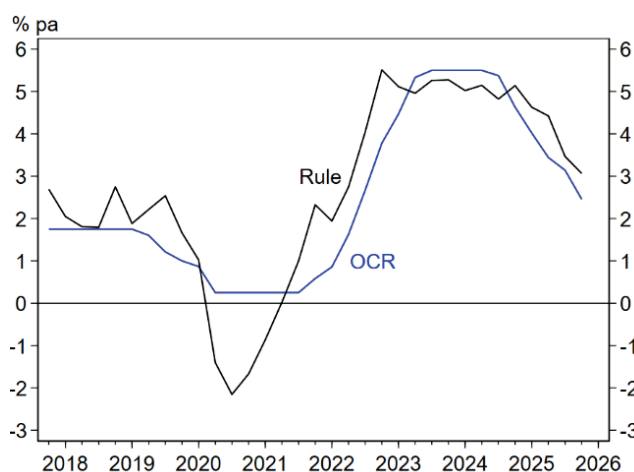
The information provided above suggests that monetary policy could potentially have been better calibrated in this episode but does not provide a clear guide on how that could have been achieved in real time. Such an assessment requires comparing alternative monetary policy strategies that could have been considered before the pandemic and their economic outcomes, had they been implemented, instead of actual policy.

One approach would be to compare the economic outcomes associated with counterfactual policies derived from simple policy rules, instead of alternative policy paths as presented in Figure 4.1. In principle, this could be done by the RBNZ using the NZSIM model, similarly to the analysis in Cherry and Kendall (2025). A narrower exercise, available to us, is to examine the real-time prescriptions of the policy rule used in the version of NZSIM that was employed for the preparation of the MPS projections.

While the policy rule embedded in the RBNZ's forecasting model does not provide a complete description of the MPC's monetary policy strategy, it is an integral part of the strategy. In the period under review, the actual OCR decision in each quarter was a discretionary choice by the MPC that reflected the MPC's implementation of its risk management framework but was also influenced by the analysis and forecasts shaped by the model and the policy rule embedded in it. In each quarter, MPC could have decided to implement the OCR prescribed by the rule, or use its discretion to deviate from that. Thus, in each quarter, the difference between the real-time rule prescription and the OCR can be interpreted as a measure of the discretion applied by the MPC at the MPS meeting—effectively capturing the part of the MPC's policy strategy for setting the current policy rate that is not adequately captured by the rule.

Figure 4.3 compares the OCR with the real-time prescriptions from the policy rule in the forecasting model. In each quarter, the prescriptions are based on the specification and parameters corresponding to the forecasting model employed for the preparation of the MPS in that quarter, using the data/projections in the MPS.

Figure 4.3 – OCR and real-time prescriptions from policy rule in forecasting model



Notes: OCR is the quarterly average of daily rates. Real-time rule prescription in each quarter uses the policy rule employed for the preparation of the MPS and corresponding data/projections.

²⁰ See Krippner (2016) and Culling et al (2019) for details on the construction and properties of EMS.

²¹ Another way to assess the degree of monetary policy easing would be compared to other central banks that faced a similar challenge in 2020. Krippner (2021), characterizes the RBNZ easing as “deepest and steepest,” even though the health and economic impact of the pandemic was less severe in New Zealand.

Two features are particularly noteworthy. First, during 2020 (starting with the May 2020 MPS that first incorporated information about the pandemic) the rule prescribed a negative setting of the OCR. Thus, once it was decided that 25 basis points was the effective lower bound for the OCR, the rule supported the activation of alternative policy measures, as implemented by the RBNZ early in the episode. Second, the rule called for an earlier start of policy normalization and faster removal of stimulus during 2021 and 2022. In 2021Q3, when the MPC contemplated a 25 basis point increase, the rule prescribed a 75 basis point hike to 1 percent. And in each quarter from 2021Q4 to 2022Q4, the rule prescribed a setting for the OCR that was over 100 basis points higher than actual policy.

Though not directly comparable, the prescriptions presented in Figure 4.3 are more in line with policy scenarios 1 and 3 in Figure 4.1 than with actual policy. This exercise suggests that if the OCR had been set in accordance with the prescription from the policy rule in the RBNZ's forecasting model, as was available in real-time, the MPC would have fulfilled its dual mandate better.

5. Vulnerability to policy mistakes

Section 2 described issues that may arise for medium-horizon IFT (and other types of flexible inflation targeting) strategies due to them being underpinned by assumptions of perfect knowledge that are at odds with the real world. Good results can come in benign circumstances, but bad results in challenging ones. Section 3 described policy errors made in 2020-22's challenging circumstances. The Review and Assessment of the Formulation and Implementation of Monetary Policy (RAFIMP) and other Bank retrospectives have identified knowledge gaps and laid out related work programmes.

Lack of robustness in challenging circumstances has thus far been explained at a high level. This section is more specific in describing systematic vulnerabilities within the RBNZ's medium-term horizon IFT approach to policymaking. Forecast errors are first documented (section 5.1), highlighting that they were not random, suggesting something systematic at work. Three types of vulnerability to systematic errors are pulled out for particular attention. The first relates to the unobservability in real time of trends in key determinants of economic dynamics (section 5.2). The second relates to the possibility of non-linear economic dynamics rather than linear relationships presumed when forecasting (section 5.3). The third relates to human factors involved in formulating views (section 5.4) and making decisions (section 5.5).

In addition to the removal of the maximum sustainable employment objective, additional modifications to the policy strategy are available, as are other ways of robustifying it. They are covered in Section 6.

5.1 Forecast errors in an Inflation Forecast Targeting context

An IFT strategy relies heavily on good forecasts to guide policy. However, forecasts are far from perfect and are worse in more turbulent times and at longer horizons. With imperfect but unbiased forecasts, using a forecast targeting "window" that corresponds with policy lags may allow effective inflation control with minimal disruption to the real economy. But if forecast errors are persistently one-sided, for a policy-relevant duration, they risk throwing the policy path off track.

Bank forecast errors during the COVID period were unusually large and were persistently in the same direction (Figure 5.1). Inflation forecast errors were larger than in earlier episodes, including the GFC. Near-term (one to four-quarter ahead) inflation forecasts undershot by more than 2.5 percentage points for forecasts made in 2020Q4 to 2021Q3. Crucially, longer-term forecasts — that are important in the New Zealand policy strategy — undershot by more. Forecast errors that were persistently in the same direction, over the period, were an important contributor to poor policy performance.

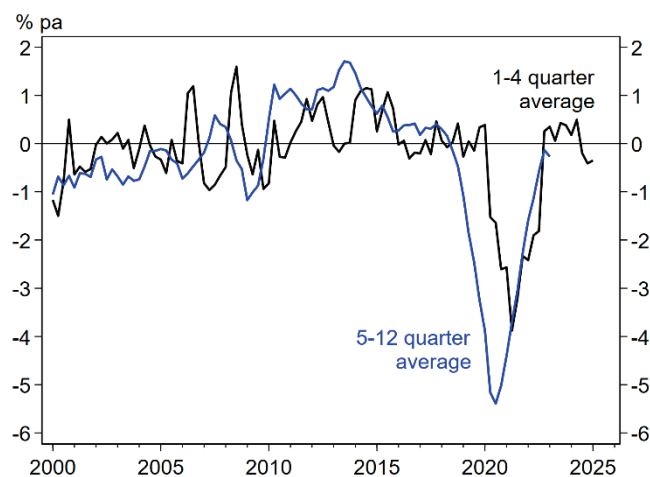
Figure 5.2 shows the succession of forecasts for the policy rate. The paths projected—the “whiskers” in the chart—are the paths that, at each forecast round, were thought needed to achieve inflation target over the medium term. The realised policy interest rates—those actually needed—are shown in blue. Errors in successive forecasts were persistently in the same direction.

5.2. Unobservable trends and misperceptions

One potential source of persistent forecast errors is unobserved trends in key variables. To explain this, it is useful to briefly review how trends influence forecasts in the RBNZ forecasting process.

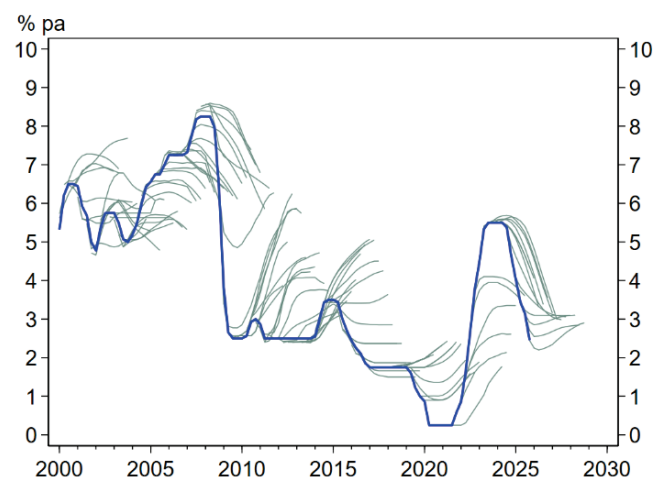
Beyond the near-term, the RBNZ heavily relies on a macroeconometric model for forecasting. During the period under review, this was NZSIM.²² Similar to earlier Bank models, NZSIM is a “gap” model. The evolution of key variables in such models can be described by decomposing them into slow-moving or static trends, often called “stars”, and cyclical “gaps” related to business cycle dynamics. For instance, potential GDP is the star variable (the trend), corresponding to real GDP while the output gap is the corresponding gap.

Figure 5.1 – Average inflation forecast error



Notes: Forecast errors are annual inflation forecasts minus outturns for each MPS.

Figure 5.2 – Interest rate forecasts

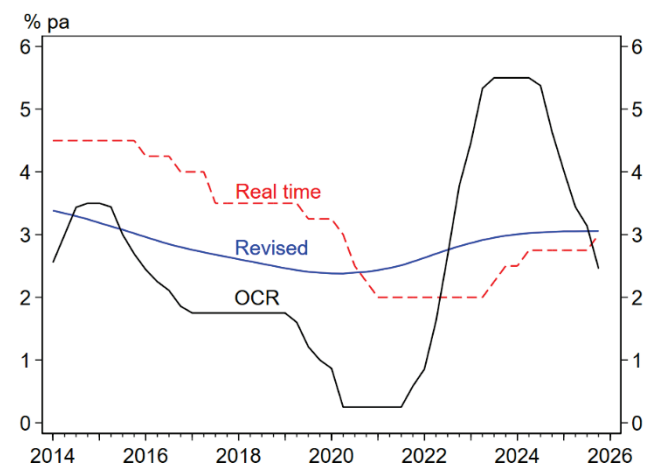


Notes: Actual OCR (blue line) with interest rate paths from each MPS. Paths are for the OCR projections (since Nov. 2016) and for the 90-day bank bill rate (minus a spread of 25 bps) for earlier MPS.

²² Richardson (2016) provides an overview of the policy process and the role of the model for forecasting beyond the near term. The structure of NZSIM is explained in Kamber et al (2015). Sander (2015) describes some key differences in the version of the model employed for forecasting.

Key star variables are potential GDP, the real neutral interest rate, r^* , and the nominal neutral interest rate, i^* . Inflation's star variable is the target. The levels of all of these, and any trend growth in them, need to be assumed. Given those assumptions, the monetary policy transmission mechanism and inflation dynamics can all be described in terms of the gaps. Reducing the policy rate so that the real interest rate is below its real neutral level (a negative real interest rate gap), raises the output gap (the difference between actual and potential output). A positive output gap raises the inflation gap (the difference between actual inflation and the inflation target). In the model, the real interest rate gap serves as the indicator of the stance of interest rate policy and the output gap serves as the critical intermediate step between the policy rate and inflation.

Figure 5.3 – Neutral rate misperceptions



Notes: Real-time series shows, in each quarter, the estimate of the nominal neutral OCR for that quarter corresponding to the MPS published in that quarter. Revised estimates as of May 2026.

Assumptions for the star variables involve generating real-time estimates of the true stars that actually influence economic behaviour. The difference between the estimates and the true value—misperceptions about stars—will translate to forecast errors.

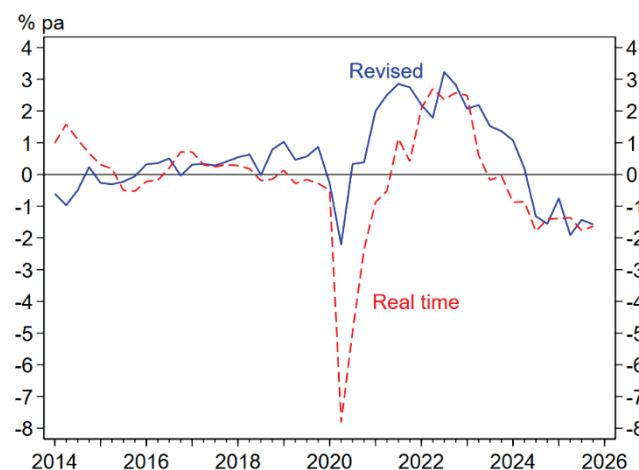
The sign and size of misperceptions can be illustrated by comparing real-time and revised RBNZ estimates of key concepts. Figure 5.3 compares these for the nominal neutral rate. Corresponding estimates of the real neutral rate can be obtained by subtracting the 2 percent inflation target from both series. The misperceptions in the real and nominal concepts are identical.

The real neutral rate, r^* , is a critical element for assessing the stance of interest rate policy. Policy is “tight” when the real short-term interest rate is higher than r^* and “easy” when it is lower than r^* . Misperceptions regarding r^* imply that policy will be sometimes tighter and sometimes easier than the bank intended.

The direction of the misperception in the neutral rate is informative for forecast errors before and during the period under review. From 2014Q1 (the start of the period shown) and until mid-2020, the real time estimates were above the current estimates in every quarter. In this period, the RBNZ was providing less stimulus than intended helping explain why inflation was below target before the pandemic. Real-time estimates of r^* were then revised surprisingly sharply downwards over a few quarters before and during the pandemic. The lowest real-time estimate was zero. From mid-2020 onwards, real time estimates were consistently below current estimates in every quarter. Based on the current estimates, the RBNZ was providing more stimulus than intended. Misperceptions about r^* was one factor explaining why inflation forecasts were persistently too optimistic in that period. In 2023, the RBNZ started revising the estimates upwards.

In a similar vein, Figure 5.4 compares current estimates of the historical output gap with its real-time counterpart, corresponding to the MPS published in the quarter. The difference between the series reflects revisions in the estimate of potential GDP as well as the nowcast error for real GDP. During the period of interest, real-time estimates of the output gap were systematically more pessimistic than is evident with current data. In the model, the output gap is the most critical link between the stance of monetary policy and inflation. Given the model's inflation dynamics, persistently pessimistic real-time assessments of the output gap meant persistently too low inflation forecasts. These misperceptions also help explain why inflation forecasts were persistently too optimistic, and why inflation forecast errors were autocorrelated in the period under review.

Figure 5.4 – Output gap misperceptions



Notes: Real-time series shows, in each quarter, the estimate of the output gap for that quarter corresponding to the MPS published in that quarter. Revised estimates as of May 2026.

Misperceptions about trend GDP are also useful for understanding delays in normalizing policy observed during the period under review. Consider the output gap and inflation data and forecasts associated with the November 2020 MPS. At the meeting, the MPC decided to provide more stimulus by introducing the Funding for Lending Programme (FLP). At the time, the output gap for the preceding quarter, 2020Q3, was estimated to be about minus 2 percent. The nowcast, for 2020Q4, was even worse and the output gap was projected to stay negative into 2022. This contributed to the inflation *forecast* staying below target despite the sharp rise observed in actual inflation in 2020Q3. In the event, as suggested by the current estimates, the output gap was already positive in 2020Q3, the quarter before the November 2020 decision. The economy was already overheated, but this was not evident in real time.

Real-time estimates of many of the star variables employed in forecasting are highly uncertain and also need to be revised over time. Unobservable trends are also very hard to model. These are well-known sources of anomalies in forecasting, especially at longer horizons.²³

5.3. Other non-linear dynamics

Misperceptions of trending star variables is just one source of non-linearity. Selected others will be noted in this section. They are relevant to forecasting and policy implementation because modelled economic processes are typically presumed to be linear in character. The model used by the Bank is no exception.

²³ Using forecasts for the United States, Farmer et al (2024) document how anomalies in forecasting arise and can be reconciled even with optimal updating of beliefs in a Bayesian setting when unobservable trends are uncertain and difficult to model.

The success of inflation targeting in dealing with chronic inflation without making the real economy unstable has largely been attributed to inflation expectations becoming “well anchored”. As explained in Section 2, that understanding underpins the medium term IFT policy strategy. In reality we know relatively little about how agents form their views about the pricing environment in which they are operating.

Imperfect observability of inflation developments requires constant learning so that inflation expectations can be kept up-to-date. In the literature, various approaches have been investigated to capture imperfections in the formation of inflation expectations and their interaction with monetary policy. One is to introduce perpetual learning.²⁴ Another is to recognize that, rather than continuously updating expectations, businesses and households are generally “inattentive” to the risks of generalised inflation when recent experience suggests there is little to worry about.²⁵ They only start paying attention after being sufficiently alerted by adverse developments in prices they encounter. Given the discontinuity in the degree of attention being paid, inflation might already have become high and volatile.²⁶

A common finding is that imperfections in the formation of inflation expectations—or more generally in agents’ perceptions of their pricing environment—introduce nonlinear dynamics that interact with monetary strategy and have first-order consequences for policy design.²⁷ The behaviour of inflation expectations in New Zealand confirms the importance of understanding and accounting for this dynamic. Evidence points to New Zealand firms and households operating inattentively in normal times.²⁸ And the RBNZ has shown that short-term inflation expectations were not well anchored post-COVID, even less than in other advanced economies (Davies, 2025, Figures 65-68). These expectations rose to the same height as realised inflation, with little lag.

It appears that the possibility of instability in inflation expectations was insufficiently considered in the policy process. Too heavy reliance was placed on the Bank’s long-term credibility acting as an ever-present stabiliser. This is one reason why the model’s and MPC’s medium-term forecasts of inflation remained benign as inflation took off. Insufficient attention to the risk of unanchored inflation expectations misled MPC as to the *current* stance of policy. Assuming a stable pricing environment allows nominal interest rates to be used as a summary statistic for the policy stance (after allowing for the neutral rate). If pricing dynamics might be unstable, the nominal policy rate can be a misleading indicator. As shown in Figure 4.2, the real policy rate fell to record lows even as the Bank believed it had stopped adding stimulus and was embarking on a tightening.

²⁴ Two seminal reference are Evans and Honkapohja (1999) and Sargent (1999).

²⁵ Candia et al (2023), Coibion et al (2018), Maćkowiak, Matějka, Wiederholt (2023) and Bayarmagnai (2023).

²⁶ Recent work in the United States has estimated a threshold when typical households and firms to start paying attention to inflation developments at 4%. Beyond this point, attention doubles and supply shocks have stronger and more persistent effects on inflation, a pattern that echoes what has been seen in New Zealand post-COVID (Přaui, 2026).

²⁷ The distortion in the variability frontier shown in Figure 2.4 is an example. Borio et al (2023) and Buckle et al (2026) provide others.

²⁸ Buckle and Carlsson (2000), Coibion et al (2018) and Bayarmagnai (2023).

The insufficient attention paid to non-linear dynamics relevant to policy decision making was also evident in a tendency to delay raising the OCR during the recovery from COVID, with explicit reference to a positive “option value of waiting” for additional information to arrive that could reduce uncertainty about the outlook. In dynamic systems, the option value of waiting is positive in some circumstances, but not all. Post-COVID, slowness to tighten was costly: the option value was negative.

5.4. Forecast and decision-making biases resulting from human factors²⁹

Forecasts are not made by machines alone. Models are useful for enforcing internal consistency, but their output is sensitive to underlying assumptions. People build and calibrate the models, provide current and historical values of model variables, and provide judgemental assumptions about future paths of conditioning variables and future shocks.

There are, accordingly, many places where forecaster beliefs concerning the future can be reflected in model-generated forecasts.

Our interviews with MPC members and staff involved in preparing forecasts during the period under review revealed the relevance of collective beliefs. Interviewees frequently referred to the risk that economic recovery would be undermined by new COVID lockdowns and related uncertainty well into 2021, a concern that remained prominent despite increasingly positive incoming data.³⁰

That beliefs play a role in forming forecasts is to be expected. What is perhaps surprising, is the persistence of one narrative despite incoming information that suggested that it might not be the correct one.³¹

The risk of groupthink is well known and was a factor in the decision to introduce a decision-making committee (see next subsection).³² It was also recognized by the Reserve Bank in the Monetary Policy Handbook (RBNZ, 2020), as was the challenge of overcoming it. There is no indication that groupthink was at play in MPC decision-making during the period under review.³³ The relevant concern identified here is different: whether institutional arrangements did enough to encourage diverse perspectives and ensure that alternative views received adequate consideration in policy deliberations.

²⁹ See Tversky and Kahneman (1981) and Akerlof (1990) for behavioural biases in decision-making.

³⁰ The negative view of the recovery biased policy towards increased stimulus, as did the adoption of a Least Regrets strategy. But the two are conceptually distinct.

³¹ MPC members may also have been partly reassured by the appearance of private sector forecasts and commentaries being in line with the Bank's, suggesting that there has been some kind of independent check on MPC's thinking. Yet it is well known that private forecasts are not independent – a “hall of mirrors” exists, for a variety of reasons (Amato et al, 2002; Morris and Shin, 2018).

³² Alan Blinder noted that “group-think is an ever-present danger that [central bank] committee structures should be designed to avoid or minimize.” See also Sibert (2006), and Blanchflower and Levin (2023).

³³ One of the reviewers, Mr Archer, does not agree with this statement.

5.5. Vulnerabilities due to institutional arrangements

The Monetary Policy Committee was constituted by an amendment to the Bank's Act at the end of 2018 and became operational in 2019. Part of the motivation was an expected reduction in the risk of a single viewpoint being brought to bear on difficult decisions.³⁴

Over the period under review, the arrangements did not encourage a sufficiently broad range of perspectives in policy deliberations. Macroeconomic forecasts have famously wide confidence intervals. Yet, in late 2020 and early 2021, the risk that too much stimulus had already been provided and inflation might take off did not feature prominently in the discussion. This raises questions about the effectiveness of arrangements intended to bring a breadth of perspective.

Two weak points in the original design of decision-making arrangements stand out. The first relates to the Charter's strong preference for consensus decision making, and the presentation of a unified view of the policy decision and outlook. The second relates to the balance of insiders and outsiders on the Committee. They are taken in turn.

A preference for consensus decisions is dominant in central banking, although the trend is now towards greater individualism both inside and out of the Committee room. One rationale that has been prominent is the perceived importance of managing expectations through signalling. That requires clarity of signalling; voting disparities and multiple perspectives being aired in public risks a "cacophony" of signals.³⁵ Yet too much store can be placed on presenting a clear, single narrative. There are extremely wide confidence intervals around any single scenario. There is a value in communicating that uncertainty and markets are well placed to digest that information

On balance, voting with attribution provides a stronger incentive for each member actively to assess the viewpoint with which they associate themselves. The introduction of attribution for votes announced in April 2026, and the revealed greater preference to put issues to a vote, are both welcome. Near real-time disclosure of members' individual reasoning would work in the same direction. MPC might consider, as possible reference points, the approach now taken at the Bank of England, where each member provides a paragraph summarising their view, for inclusion in meeting minutes; the Sveriges Riksbank approach, where Executive Board members are able to have their reasoning fully expressed in the meeting minutes (which are typically published with only a five-day lag); and the New Zealand Supreme Court's practice of publishing both dissenting and concurring fully-reasoned opinions when differences in reasoning are substantive.

The second weak point in decision making arrangements concerns the balance of the committee as between members who are employed by the Bank (and so qualify as potential internal members) and report to the Governor, on the one hand, and part-time externals on the other. Internal members who report to the Governor are, for a variety of incentive reasons, more likely to align their viewpoint with the Chair's. There is no indication that this weakness was a factor in MPC decisions during the period under review, but it presents a risk that may well be a factor in future.

³⁴ Rennie Report (2018).

³⁵ Blinder (2018), Issing (2005).

6. Robustifying the framework³⁶

6.1. Shortening the policy horizon

The foregoing discussion has highlighted vulnerabilities in the IFT strategy. A common theme is imperfect knowledge, the significant possibility of model misspecification and non-linearities, and the medium-term horizon. Forecasts beyond three to four quarters contain little useful information. Of more significance, targeting a longer horizon permits systematic errors to grow to a material size.³⁷

Section 5 argued that forecasts are in practice prone to systematic errors, especially in unusual and unstable conditions, and especially at longer horizons. Generally speaking, the calibration of policy horizons used for IFT has aligned the policy horizon with monetary policy's peak impact because that promises good stabilisation results.³⁸ But those good outcomes are most assured in a perfect information environment. Allowing for uncertainty, robustness analysis argues in favour of shorter horizons to reduce the risks arising from model misspecification and other manifestations of imperfect knowledge.³⁹ On the other hand, shortening the IFT horizon excessively would result in unnecessary economic volatility.

A priori, it is difficult if not impossible to assess the balance between possible costs and gains at different horizons. Making that assessment requires model-based robustness analysis that allows for plausible forms of model misspecification, measurement errors and possible non-linearities. The work needed to identify a suitably robust horizon is not simple, but there is a *prima facie* case that improved robustness would result.

6.2. Scenario analysis

Scenario analysis is another important tool for robustifying policy strategy. It can serve multiple purposes spanning MPC's quantitative risk analysis and policy communications.⁴⁰

The MPC decision to reduce systematic scenario analysis, starting with the May 2019 MPS, was unfortunate.⁴¹ During the pandemic, the absence of systematic scenario analysis contributed to the domination of untested narratives at MPC meetings, that proved particularly costly. Systematic scenario analysis would have given individual MPC members a tool and the information useful for questioning the dominant narrative, prompting better risk analysis and, in turn, delivering better decisions. The recent increased focus on scenario analysis is a welcome step.

³⁶ The suggestions offered in this section presume that an IFT framework remains at the core of MPC's policy strategy. We do not consider state-contingent strategy design, as in Harberis et al, (2025),

³⁷ The horizon in the period under review can be inferred from the 2020 edition of the Monetary Policy Handbook: "Historically, the RBNZ's monetary policy strategy has generally been to choose a policy stance that is expected to result in inflation reaching the target mid-point after about two years." (RBNZ, 2020, p. 67)

³⁸ See e.g. Batini and Haldane (1999), Batini and Nelson (2001) and Drew and Orr (1999).

³⁹ See e.g. Woodford (2007) and Levin et al (1999, 2003).

⁴⁰ Numerous authors have advocated a role for scenario analysis in monetary policy. Recent work includes Archer and Levin (2018), Orphanides (2019), Bordo, Levin, and Levy (2020), and Bernanke (2024).

⁴¹ In the period under review, alternative scenarios, used in a systematic and disciplined way to illuminate the nature of risks, were absent not only from most MPSs but also from the briefing material informing MPC deliberations in many meetings.

Robust policymaking with an Inflation Forecast Targeting strategy requires careful analysis of the uncertainty and risks surrounding the baseline projection, especially in an uncertain environment. A systematic and disciplined use of scenarios is an essential part of careful analysis. Disciplined use requires that scenarios should be sufficiently broad to examine key assumptions and judgements, and to cover a material proportion of upside and downside risks to the outlook in real time. Deployed that way, scenarios can be especially useful for evaluating and explaining the potential impact of plausible, but low probability, events that may have asymmetric consequences sufficient to meaningfully influence the setting of monetary policy. Their disciplined use serves as a robust defense against groupthink.

When policy settings are influenced by risks, the public presentation of scenarios helps provide a fuller rationale for policy than is available from the publication of a single baseline scenario or forecast. The May 2020 MPS was more informative, in this regard, than subsequent MPSs published in the period under review.

Scenarios also have an important use in designing MPC policy strategy. They can aid the evaluation of key elements, comparing systematically the performance of alternative strategies over history. Consider, for example, the MPC's choice of target horizon. The implications of different horizons can be quantified with counterfactual simulations using real-time data, and the analysis extended to evaluate historical performance with alternative target horizons over a longer period. Historical analysis along these lines would uncover the vulnerabilities discussed in the previous section, helping the MPC decide how the target horizon should be adjusted to ensure better outcomes in future.

6.3. Simple rules

Simple rules linking the policy interest rate in the current quarter to a small number of macroeconomic variables can be used in two ways. In one, simple robust rules can serve as a cross-check on the MPC's discretionary decision for the interest rate in the current quarter, guarding against unavoidable misspecification in the dominant model or process. In the other, a simple rule can serve as a benchmark policy rule embedded in a model-based forecasting and policy process and serve as a starting point for the policy decision. This helps bound the application of discretion and helps ensure that policy remains systematic over time. Both dimensions are relevant to RBNZ practice in the period under review.

Simple robust rules are one way to address vulnerabilities in monetary policy.⁴² Simple robust rules typically recommend a setting for the OCR on the basis of just one or two macroeconomic indicators, such as nowcasts/near-term forecasts of inflation and economic activity. They are not meant to be "optimal," when set within the complex dynamic structure of the economy, nor within any specific model that may be employed for standard forecasting and policy analysis. They are meant to be robust, protecting against vulnerabilities such as those reviewed in section 5.

⁴² See Taylor and Williams (2011) for a review.

Jan Qvigstad (2006), then Chief Economist at the Norges Bank, offers an example of how to employ simple rules as a cross-check in the context of IFT. He suggested a set of criteria for implementing inflation targeting, including five describing the standard forecasting and policy analysis process, similar to RBNZ practice. To address the vulnerability of this practice, he included the following criterion:

“It may be useful to cross-check by assessing interest rate setting in the light of some simple monetary policy rules without using the model framework of the first five criteria. If the interest rate deviates systematically and substantially from simple rules, it should be possible to explain the reasons for this.” (Qvigstad, 2006, p. 11.)

Applying the Qvigstad criterion can help guard against major mistakes. Simple robust rules cannot capture all the elements of the projection but can provide a warning if the recommended policy in the current quarter is notably out of line with what is known to be more robust. Public disclosure of prescriptions from one or two simple robust policy rules in the MPS would also help inform the public of MPC’s treatment of risks, and aid accountability. These policy prescriptions would be presented in the MPS for the current and next quarter.

The selection of simple robust rules would draw on policy research specifically relevant to the New Zealand context, as well as past and ongoing research in the global central banking community.⁴³

Policy research exploring robustness characteristics of simple rules would also be useful to help the MPC agree on the specification of a benchmark policy rule embedded in the RBNZ’s forecasting model. MPC can use this benchmark rule as a starting point for the decision on the policy rate for the current quarter.

During the period we explored, the Reserve Bank’s forecasting model included a simple policy rule. The specification of the policy rule is critical for the dynamic properties of the model and the transmission of monetary policy and is thus an integral part of policy strategy. As discussed in section 4, the setting for the OCR in each quarter reflects discretionary decisions. During the period we explored, the current-quarter OCR deviated, at times notably, from the prescription corresponding to the policy rule in the forecasting model (Figure 4.3).

We recommend that the MPC agree on the specification of this policy rule for reasons of consistency in the preparation of forecasts and greater transparency on discretionary adjustments to policy.

Policy research can help the MPC adopt a benchmark policy rule in two ways. First, with historical analysis, capturing characteristics of policy in periods when policy was successful.⁴⁴ Second, with a rigorous evaluation process aiming to identify properties that contribute to robustness, taking into account multiple dimensions of uncertainty that can be incorporated into policy evaluation.

⁴³ For example, in other economies, rules that respond to nominal income growth have been identified as useful. In the New Zealand context, McKibbin (2022) observed that GNE growth would have been more useful than GDP growth. Examining vintages of near-term projections of a proxy for GNE, provided by the Bank, we agree that monitoring GNE would likely have helped the Bank in real time during the period under review.

⁴⁴ Estimated rules alone cannot achieve this as historical episodes invariably include periods where policy has been problematic.

Among others, this should include uncertainty about trends, about the structure of the economy, about expectations formation, in essence vulnerabilities along the lines discussed in section 5.⁴⁵

The adopted benchmark policy rule should be transparently communicated to help make policy more systematic. A benchmark would serve as a useful input to MPC, to constrain but not eliminate discretion. The benchmark policy rule would require periodic review and adaptation. MPC could require periodic reevaluation of models and forecasting procedures, including the benchmark policy rule, and transparently communicate updates.

7. Alternative Monetary Policy (AMP) Tools

We are asked (Terms of Reference 12.4) for our views on the MPC's and the Reserve Bank's choices on the design and use of monetary policy tools. Terms of Reference 10.2 also seeks recommendations to improve the monetary policy response to future shocks.

Global evidence and analysis supports the use of AMP instruments, including **Quantitative Easing via LSAPs**, in situations where the ELB has been reached and additional monetary stimulus is needed.⁴⁶ LSAPs can also be used to help solve problems with bond market functioning.

In New Zealand in 2020, LSAPs were used for both these purposes. The initial intervention to correct bond market dysfunction was appropriate and successful. Their subsequent use for monetary stimulus is more difficult to evaluate. Had the stimulus turned out to have been needed, the global evidence and analyses referred to suggest it would likely have been net positive with respect to income and employment and (as a consequence) also the public finances. This assessment allows for transfers from the public sector (via the central bank) to secondary market sellers of government bonds associated with the RBNZ's willingness to buy at unsustainably high prices⁴⁷. But stimulus from LSAPs turned out to contribute to excess demand, inflation and subsequently a costly economic slowdown and unemployment. On the basis of the Bank's reasonable real time expectations, the ex ante answer to the Terms of Reference's question of whether the benefits of these tools justified the risks and costs is probably positive (subject to a qualification discussed in the next paragraph). Later, however, the answer turned negative.

LSAPs were employed to a greater extent than they would have been had a zero or **negative OCR** been used first. Earlier Reserve Bank work concluded that in circumstances where Quantitative Easing (QE) might be used a reduction in the OCR to -0.75% or even -1% OCR "would be probably the most effective and simplest" response (Orr, 2019). In the end, in March 2020 MPC was advised that +0.25 percent was the lower limit for the OCR as banks had indicated that they were not operationally ready for negative rates.

⁴⁵ The existing literature is typically based on analysis of other economies but has lessons that likely carry over to the New Zealand context. For example, Taylor and Williams (2011) report that nowcasts/short-term forecasts that serve as inputs should be at most 3-4 quarters ahead, that annual forecasts of inflation are preferred over quarterly inflation forecasts, that first difference rules protect better than level rules against neutral rate and output gap uncertainty, and so on.

⁴⁶ Bernanke (2020); Hofmann et al (2025); CGFS (2019); Kolasa, Laséen, Lindé (2025).

⁴⁷ That prices were unsustainably high is inherent in the use of the instrument for countercyclical reasons. See Goodfriend (2014) and Cecchetti and Hilscher (2024).

The lack of readiness for a negative OCR was unfortunate, as examined elsewhere (NZ Parliament, 2020). The MPC's acceptance of the banks' lack of readiness, and the decision not to reduce the OCR to zero and to commit to leaving it at 0.25% "for at least 12 months", contributed to the costs of the LSAP. A lower OCR would have meant both lower funding costs for the LSAP, and considerably smaller LSAP volume, on the order of \$15 billion per 25-basis-point OCR reduction (RBNZ 2022).

This use of **unconditional calendar-based Forward Guidance (FG)**—to be contrasted with guidance made conditional on prevailing circumstances—was of questionable merit. Calendar-based FG can become a trap (Orphanides, 2024). Moreover, locking in an OCR of 0.25% for an entire year ruled out adopting a negative interest rate policy even were it found to be implementable.

Limited RBNZ work in the lead up to the COVID period had not favoured the use of Yield Curve Control (YCC) or Yield Curve Targeting (YCT). During the COVID-19 episode, the Bank of Japan and Reserve Bank of Australia employed these tools with success but only the Bank of Japan managed the exit well. The Reserve Bank examined the RBA approach in May 2020 but by then LSAPs had already pushed NZ government bond yields below the OCR, so there was not much to be achieved by providing a clearer statement of the Bank's operational objective for the yield curve.

LSAPs and YCC/YCT are two routes to the same end: a compression of longer-term rates, when a reduction in the OCR is not possible. Whereas LSAPs target a quantity of purchases with an interest rate result in mind, YCC/YCT target a point or a region on the yield curve, expecting to need a smaller quantity of purchases—and so financial risk—to achieve the same outcome (but at the risk of needing more).

The analysis given to MPC argued that exit from such policies could be more difficult than exit from LSAPs. Indeed, the RBA encountered significant difficulties in exiting from YCT, but this reflected mishandling of the RBA's activation of LSAPs after the activation of YCT (Orphanides, 2023). YCC/YCT, coupled with state-contingent forward guidance, are useful tools that should remain under consideration for potential adoption at the ELB.

The **Funding for Lending (FLP)** instrument provides another means of lowering interest rates through both OCR funding and lowering other funding rates. Since bank funding markets were functioning well, the merits of the FLP were doubtful even at the time. At the MPC meeting where the decision was made to implement FLP, MPC received advice that the transmission mechanism was functioning unimpeded, and that banks had access to ample funding. According to our interviews, banks were surprised that the FLP option was pursued but were happy to substitute cheap RBNZ funding for more expensive alternatives. In general, credit support instruments risk distorting credit allocation, whether that be through the selection of eligible intermediaries, rules on access, or impacts on credit pricing differentials.

As a final and more general comment, the Bank should periodically review these alternatives and ensure their operational readiness for implementation in an evolving financial landscape.

8. Monetary-fiscal policy interaction

Paragraph 12.5 of the Terms of Reference asks about the fiscal/monetary boundary. The COVID crisis led to monetary and fiscal policies interacting in ways not seen in recent history, raising questions about future interactions at or near the effective lower bound. Income support and other targeted policies acted countercyclically by providing stimulus to limit the severity of the downturn. The Bank engaged in LSAPs, initially to correct government debt market malfunctioning—an interest of the Treasury’s debt management operation—then subsequently for macroeconomic stimulation. In the latter role, LSAPs created fiscal risks while reducing the government’s debt raising costs and supporting outcomes that would result in fiscal gains. The speed and scale of fiscal actions were highly relevant to the macroeconomy’s outlook as well as to term interest rates, impacting the Bank’s calibration of its policies.

During the initial COVID response phase, fiscal and monetary policies were pulling in the same direction without actively being coordinated: coordination remained appropriately “at arm’s length” as each authority pursued its respective objectives, based on shared information. But rather than the Bank having time to plan its response to fiscal policy (in normal times, there is an annual budget cycle and delayed implementation), it needed to make rapid assessments of the extent to which just-announced actions would affect the macroeconomy and when.

The combined stimulus ended up being more than was needed, largely as a result of greater than expected economic resilience and the surprising effectiveness of various policy actions, not because of a failure to actively plan or coordinate policies, or to provide adequate information. With respect to the latter, information flowed impressively freely at staff levels. Supplementary information provided direct to MPC via the Treasury observer was initially variable, but became more standardised over time.

Indeed, active planning for the coordination of policies could risk fiscal dominance. Such risks are the prime reason for institutional separation, the assignment of distinct objectives, and the independence of the central bank. The COVID experience does not lend support to a change in institutional arrangements. Arm’s length coordination remained an effective institutional arrangement; any mistakes in calibrating policies as occurred would not have been avoided with joint planning or implementation.

With respect to quasi-fiscal instruments that might be used by the Bank—such as LSAPs and FLP—there is currently no protocol that governs their availability for monetary policy purposes.⁴⁸ It seems sensible to consider the range of non-standard instruments available to the Bank holistically and in peacetime. This would support legitimacy and help ensure the Bank has the tools needed in exceptional circumstances.

⁴⁸ In contrast, protocols are in place with respect to the Bank’s ability to intervene in foreign exchange markets.

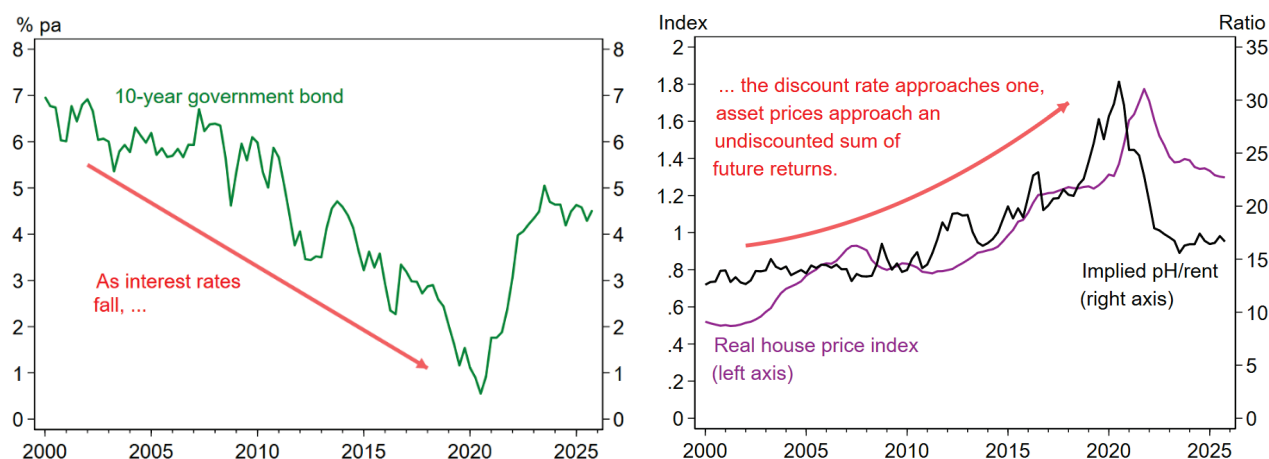
A contributor to the Bank's continuation of (what turned out to be) unneeded stimulus was its longstanding practice of waiting for fiscal decisions to be announced before taking them into account. The Bank is quite reasonably averse to being seen to be "frontrunning" fiscal decisions. But there are ways of allowing for a likely fiscal policy reaction, on the basis of well-understood public fiscal policy objectives, without implying specific knowledge of government plans. In an Inflation Forecast Targeting framework the likely fiscal policy reaction to future developments and risks should be incorporated to avoid biased inflation forecasts. For example, MPC was concerned about the risk of widespread labour market dislocation following the ending of the wage subsidy scheme, without adequately considering the additional fiscal support that would be provided if needed. As a result, more monetary stimulus was seen to be required. Endogenous fiscal responses to economic developments can be allowed for in a number of ways, such as including estimated rules in the model, or including rules calibrated to automatic stabilisers, and/or employing scenarios consistent with fiscal objectives.

9. Housing

In 12.6, we are asked to review the MPC's consideration of the interaction of monetary policy with fiscal and macroprudential policy, and the impact on asset and house prices.

Several extraordinary factors affected housing in the period under review. At the onset of COVID, there was concern about banks' ability to continue to lend through the crisis. With incomes expected to fall sharply, it was anticipated that banks would need to provide payment deferrals and loan restructuring. There was also a concern that house prices might fall, as they did during the Global Financial Crisis, reinforced by a sharp fall in net immigration. The Bank (not the MPC), employing its macroprudential authority, removed LVRs for a year. Though not a monetary policy decision, under the circumstances that eventuated the opportunity to increase leverage contributed to the housing market boom and aggregate demand. In the event, unlike the GFC, house prices rose sharply, supported by Covid-induced changes in housing preferences (for example, demand for working-from-home space), LVR easing, very large fiscal stimulus, and the exceptionally low mortgage interest rates delivered by monetary policy.

The non-linear relationship between asset prices and long-term interest rates is often under-appreciated because it is less important in normal times but becomes more so as longer-term interest rates go to zero and negative values. Asset prices are more sensitive to long-term interest rates, when rates are very low (see Kocherlakota 2013, among others). This can be seen in a simple asset price model, where a house value is a discounted sum of future housing (rental) services, over 50 years. Applying the idea to New Zealand data, as an illustrative rather than empirical exercise (many other factors influence asset prices), and using the 10-year government bond (Figure 9.1, left panel) as a proxy for long-term rates, the implied house-price-to-rent ratio is shown in the right panel of Figure 9.1.

Figure 9.1 – Interest rates and house prices

Source: RBNZ interest rate data. Corelogic house price index. Stats NZ CPI. Notes: The house price to rent ratio is proxied by the house price index divided by the CPI, rescaled. The asset price-implied house price to rent ratio is calculated as $(1-\beta)T/(1-\beta)$, where $\beta=1/(1+r)$, and r is the 10y government bond plus a 3.5% proxy for the mortgage premium, net of 2% inflation.

The Bank's analysis included the interactions between monetary policy and housing. Lower interest rates, all else equal, inflate housing prices and feed into consumption. That is a normal part of the transmission of monetary stimulus. Shorter interest rates have a powerful effect on borrowing costs. Asset prices, including house prices, are sensitive to interest rates. But the increasingly nonlinear relationship with long-term interest rates at very low interest rates was likely an underappreciated driving factor in real time, in the second half of 2020.

Asset prices are both volatile and difficult to forecast, making scenario analysis a useful way of exploring and testing assumptions.

10. Other institutional issues

Certain institutional issues that may have affected MPC decision making, during the period under review were covered in Section 5. This section addresses others.

10.1. Remit updating

A core theme of this report is that the Bank's monetary policy strategy became increasingly flexible with respect to its pursuit of inflation stability starting around the adoption of IFT and running up to the introduction of the Dual Mandate. Ultimately that flexibility facilitated the large policy mistakes experienced in the period under review.

One aspect of increasing flexibility relates to the early departure from the 1989 Act's vision of Policy Targets Agreements (PTAs). PTAs—now Remits—were intended to be *for the term of office of a Governor*, absent exceptional circumstances. They were there to clarify operational objectives, consistent with the statutory objectives, in the form of an observable and measurable target or targets, facilitating a clarity of purpose and accountability. The essential example is translating “stability in the general level of prices”—the statutory objective for the primary function—into a numerical inflation target established for the term of office of the Governor.

Starting as early as 1996, practice moved away from linking PTAs to the term of office of the Governor. New PTAs started to be negotiated on the appointment of a new Minister of Finance. Rather than disconnecting operational objectives from electoral timetables, PTAs became more reflective of a new Minister's preferences—at least within the limits of what the Governor of the day was prepared to agree is consistent with the statutory objectives. That directional change became even stronger when new Remit arrangements were introduced in 2018, regrettably changing its status from an agreement, where the Governor had a veto, to a Ministerial direction subject to having obtained MPC advice.

The increasing alignment of operational objectives with electoral cycles provides additional scope for frequent and disruptive changes in the policy strategy. A time dependent Remit revision process is more sensible, and a useful move was made in that direction in the 2021 RBNZ Act with a requirement for Bank advice on the Remit to be provided at least every five years. But the Minister is not constrained from altering the Remit at any time. Stronger restrictions on the periodicity of Remit changes would be sensible, as recommended in the Rennie Review, IMF (2025) and OECD (2026). The Canadian revision process is especially commendable.

10.2. A financial constraint on policy options

Amendments to the RBNZ Act in 2019 introduced two forms of financial constraint on the MPC's ability to set monetary policy consistent with its assessment of what is needed to achieve statutory objectives. Although neither of those financial constraints was in force during the period under review, for the benefit of ensuring that monetary policy has the necessary capacity in future such episodes, it is worth asking what would have happened if either had been in force.

The two constraints are as follows.

Section 208 of the Act provides that the Minister of Finance may direct the Reserve Bank to take all reasonable steps to maintain a minimum level of capital determined by the Minister and have regard to the Minister's expectations as to the Bank's financial risk management. There is no carve out for financial risks associated with monetary policy operations. This makes those operations subject to any directive issued under s.208.

Section 47 makes the Bank's Board responsible for the Bank's financial management, and Section 121 states that the Bank is *not required* to implement a monetary policy formulated by MPC if it judges that to be inconsistent with its financial management obligations. (Elsewhere in the Act, it is made clear that the MPC formulates monetary policy but it is the Bank that implements that policy.)

Accordingly, the Minister or Board may decide that financial risks to the Bank—thence the Crown—shall override the Bank's monetary policy objectives. There are clear problems with this arrangement.

With respect to the Minister's powers, the Act does not mandate that the use of s.208 be consistent with the Remit the Minister has given to the Bank. This is in contrast with the Minister's s.125 power to redirect monetary policy towards a different economic objective. In such cases the Act requires that the Minister reconsider the Remit and adjust it if necessary. There is no such consistency check on the use of s.208.

There are some checks on a Minister's concern for financial risk dominating wider economic objectives. The Minister has responsibilities for the economy as well as for the Crown's finances. If the Bank were prevented from using its instruments to fight a deflationary slump, the Minister would be obliged to use others—primarily fiscal—at their disposal, and be politically accountable. But it remains possible for a Minister to see s.208 as providing a lever that could be used tactically for electoral advantage.

With respect to the Board's powers to refuse to implement the MPC's monetary policy on the grounds of risk to the Bank's finances, the design flaws in the legislation are more obvious. The Board could stymie monetary policy in an endeavour to have the Bank "operate in a financially responsible manner", with no legal responsibility or structural ability to take Crown finances or the overall health of the economy into account.

The fixes will require consideration of several factors that do not tidily come together, and we do not offer solutions. However, it is advisable to seek solutions that are implementable well before the Bank is required by unfortunate circumstances to take on large financial risks to protect the New Zealand economy from a deflationary slump.

Annex 1: Timeline of economic news, other developments and monetary policy decisions⁴⁹

NEWS		RBNZ DECISIONS (ex RBNZ, MPC, Other RBNZ)
		20 Dec 2018: RBNZ Act 2018 amendment
	2019 Q1	
Q4 inflation: 0.4%q/q (% pa); 1.9%y/y Q4 Unemployment 4.3% Q4 GDP: 0.6%q/q; 2.8%y/y	▶	February 2019: MPC Charter, Code of Conduct and Remit published
	2019 Q2	
Q1 inflation: 0.4%q/q (% pa); 1.5%y/y Q1 Unemployment 4.2%	▶	1 April 2019: New Framework in effect: MPC appointed; Handbook published
Q1 GDP: 0.6%q/q; 2.7%y/y	▶	8 May 2019 MPC: First Meeting: OCR cut 25bp to 1.5%, <i>Inflation below target, but projected to rise only slowly. Employment near MSE. MP Handbook published in Box A.</i>
		26 June 2019: MPC Meeting OCR unchanged
	2019 Q3	
Q2 inflation: 2.3%q/q (% pa); 1.7%y/y Q2 Unemployment 4.3%	▶	7 August 2019 MPC: OCR cut by 50bp to 1% MPC Strategy Statement
Q2 GDP: 0.5%q/q; 2.4%y/y	▶	25 Sept 2019: OCR unchanged
	2019 Q4	
Q3 inflation: 2.7%q/q (% pa); 1.5%y/y Q3 Unemployment 4.2% Q3 GDP: 0.7% q/q; 2.7% y/y	▶	13 Nov MPC: OCR unchanged. <i>Growth has continued to slow. Inflation 1.5%, expected to rise slowly. Labour market tight, near MSE.</i>
	2020 Q1	
Q4 inflation: 1.9%q/q (% pa); 1.9%y/y Q4 Unemployment 4.0%	▶	12 February 2020 MPC: OCR held at 1% <i>Slow global growth; strong export prices. Inflation stable, just below target. Labour market close to full employment.</i> 10 March 2020 Speech: Principles on using unconventional monetary policy

⁴⁹ Source: RBNZ historical forecasts. Notes: Last forecast shows the latest RBNZ MPS projection prior to the release of the data shown. GDP is annual average percent change; CPI inflation is annual % change; and unemployment is % of the labour force. GDP growth outturns shown may be a rounding error out from StatsNZ numbers.

NEWS	RBNZ DECISIONS
(ex RBNZ, MPC, Other RBNZ)	
First NZ COVID case WHO declares public health emergency of international concern WHO declares pandemic	16 March 2020 Special MPC Meeting: OCR cut to 0.25% for 12 months. <i>Outlook deteriorated significantly. COVID expected to lower inflation and disrupt labour market.</i> Bank Capital increase deferred for a year
\$50b Fiscal support package Borders closed to non-residents Q4 GDP: 0.5% q/q; 2.3% y/y Alert Level 2	23 March 2020 Special MPC Meeting: LSAP up to \$30b. Increasingly severe economic situation. Financial market disruption. Inflation and employment expected to fall below target in the near term.
Alert Level 3 Alert Level 4	
2020 Q2	
Q1 inflation: 3.1%q/q (% pa); 2.5%y/y Alert Level 3	7 April 2020 MPC: LSAP: +\$3b local govt bonds. LVR limits removed for one year, effective 1 May.
Q1 Unemployment 4.2% Alert Level 2 Fiscal: \$50b Covid-19 support package	1 May 2020 MPC: LSAP up to \$60b, include IIB. <i>Businesses closing, global financial markets volatile. Wages and working hours cut, workers have been laid off.</i>
Alert Level 1 Q1 GDP: -1.6% q/q; 1.5% y/y	24 June 2020 MPC: continue stimulus. <i>Improvements in the outlook (shorter Alert 2, bigger fiscal, restrictions easing abroad).</i>
2020 Q3	
Q2 inflation: -1.9%q/q (% pa); 1.5%y/y Q2 Unemployment 4.0% Alert Level 2 (3 for Auckland)	12 August 2020 MPC: LSAP expanded to \$100b. Activity rebounded as restrictions eased. Uncertainty high. <i>Inflation likely to remain subdued. Further job losses likely</i>
Auckland reduced to Alert 2.5 Q2 GDP: -12.2% q/q; -2.0% y/y Alert Level 1 (Auckland 2)	23 September MPC: continue to provide stimulus. Preparing for negative OCR and FLP. <i>June GDP contraction large, but smaller than expected. Housing market strength.</i>
2020 Q4	
Alert Level 1 Q3 inflation: 2.7%q/q (% pa); 1.4%y/y Q3 Unemployment 5.3%	11 November 2020 MPC: FLP: December 2020. Negative OCR by year-end. <i>Economy more resilient than expected. Wage subsidy prevented job losses, but more coming. Stimulatory policy for a long time, but risks less skewed.</i>
Q3 GDP: 14.0% q/q; -2.2% y/y	Bank Capital increase deferred a further year

NEWS	RBNZ DECISIONS
(ex RBNZ, MPC, Other RBNZ)	
2021 Q1	
Q4 inflation: 1.9%q/q (% pa); 1.4%y/y Q4 Unemployment 4.9% Alert Level 2 (Auckland 3) Covid vaccination programme begins Alert level 1	LVR restrictions reinstated, 1 March. 24 February 2021 MPC: Maintain 0.25% OCR. <i>Economic rebound stronger than expected. One-off inflation pressures. Can provide additional stimulus if needed. Pent-up housing demand.</i>
Alert Level 2 (Auckland 3) Alert Level 1 (Auckland 2)	
2021 Q2	
Q4 GDP: -1.0% q/q; -2.9% y/y Australia border to re-open 19 April Alert Level 1; 86% of front line workers ≥ one dose	14 April 2021 MPC: maintain stimulus. <i>Global outlook improving. Local economy slowed. Near term price pressures (oil, supply chain)</i>
Q1 inflation: 3.4%q/q (% pa); 1.5%y/y Q1 Unemployment 4.7% Q1 GDP: 1.6% q/q; -2.3% y/y LVR investor limits tightened \$1b Māori support fund	Higher LVRs for investors, 1 May (announced Feb 2021) 26 May 2021 MPC: Maintain stimulus. <i>Strong house construction; Australia-NZ border opening. Temporary supply chain cost pressures but medium-term inflation below target. Employment recovering.</i>
2021 Q3	
Q2 inflation: 5.3%q/q (% pa); 3.3%y/y Q2 Unemployment 4.0%	14 July 2021 MPC: Reduce stimulus by halting LSAP purchases. Local economy robust, global economy improving. Expect one-off spikes in inflation in June, Sept 2021. Price pressures building.
37% of population received ≥ one dose Alert Level 4	18 August 2021 MPC: Maintain stimulus, in context of Alert Level 4. <i>Economic rebound much stronger than expected. At or above MSE. Capacity pressures, but inflation expectations anchored. One off inflation pressures. House prices unsustainable but expected to slow.</i>
Alert Level 3-4 Alert Level 2-4 Q2 GDP: 2.8% q/q; 5.1% y/y Alert Level 2-3	LVR restrictions on owner occupiers tightened
2021 Q4	
Q3 inflation: 9.2%q/q (% pa); 4.9%y/y 75% of eligible population fully vaccinated 89% of people ≥ one dose Q3 Unemployment 3.4% 95% of Aucklanders ≥ one dose	16 October 2021 MPC: OCR +25bp, to 0.5%. <i>CPI inflation expected to reach 4% in the near term. Global inflation is rising.</i>

NEWS	RBNZ DECISIONS
(ex RBNZ, MPC, Other RBNZ)	
Q3 GDP: -3.7% q/q; 5.0% y/y Alert Levels discontinued 90% of eligible people vaccinated	24 November 2021 MPC: OCR +25bp to 0.75%. <i>Economy strong, but Covid risks still present. Inflation above target. Unemployment lowest since 2007, wage pressures. Need further tightening.</i>
2022 Q1	
Q4 inflation: 5.9%q/q (% pa); 5.9%y/y Q4 Unemployment 3.2%	Housing added to MPC Remit, effective March 2022
Russia invades Ukraine Q4 GDP: 3.0% q/q; 5.6% y/y	23 February 2022 MPC: OCR +25bp to 1.0%; gradual decline in LSAP. <i>Inflation well above target Strong demand, rising global and local inflation pressures. Employment > MSE. 50bp option discussed.</i>
2022 Q2	
Q1 inflation: 7.3%q/q (% pa); 6.9%y/y Border restrictions eased Q1 Unemployment 3.2%	13 April 2022 MPC: OCR+50bp to 1.5%. <i>Demand robust, resource constraints Rising global inflation pressures. Core inflation at or above 3%. Inflation above MSE.</i>
Q1 GDP: -0.2% q/q; 5.1% y/y	25 May 2022 MPC: OCR +50bp to 2%. Continue to tighten. <i>Inflation above target. Employment > MSE. QSBO: labour constraint. Least regrets shifts to too little too late.</i>
2022 Q3	
Q2 inflation: 6.8%q/q (% pa); 7.3%y/y Q2 Unemployment 3.3%	15 July 2022 MPC: OCR +50bp to 2.5%, continue to tighten. <i>Labour shortages. Core inflation around 4%. Growing risks to medium term activity.</i>
Q2 GDP: 1.7% q/q; 1.0% y/y	17 August 2022 MPC: OCR +50bp to 3%, continue to tighten. <i>Demand resilient, Inflation 7.3%, Employment > MSE.</i>
2022 Q4	
Q3 inflation: 8.9%q/q (% pa); 7.2%y/y Q3 Unemployment 3.3%	5 October 2022 MPC: OCR 3.5%, continue to tighten
Q3 GDP: 2.0% q/q; 2.7% y/y	23 November 2022: MPC: OCR +75bp to 4.25%, continue to tighten. <i>Inflation too high. Employment > MSE. Global outlook weakening.</i>

NEWS	RBNZ DECISIONS	
	(ex RBNZ, MPC, Other RBNZ)	
	2023 Q1	
Q4 inflation: 5.9%q/q (% pa); 7.2%y/y Q4 Unemployment 3.4%		22 Feb 2023: MPC OCR+50bp to 4.75% Demand robust, signs of easing. Inflation widespread, too high. Labour market extremely tight.
Q4 GDP: -0.6% q/q; 2.4% y/y Q4 inflation: 6.0%q/q (% pa); 6.7%y/y		5 April 2023: OCR +50bp to 5.25%
Q1 Unemployment 3.4% WHO: COVID-19 no longer a public health emergency of international concern		24 May 2023: MPC OCR + 25bp to 5.5% Demand slowing. Inflation eased but to high. Tight labour market, but easing.

Annex 2: Reviewer biographical information

Athanasios Orphanides

Athanasios Orphanides is currently a professor of the practice of global economics and management at the Massachusetts Institute of Technology (MIT). He was Governor of the Central Bank of Cyprus from 2007 to 2012, and was a member of the Governing Council of the European Central Bank, responsible for formulating monetary policy for the euro area, between 2008 and 2012. He previously worked as a senior adviser to the US Federal Reserve Board of Governors and holds a PhD in economics from MIT. Dr Orphanides has reviewed monetary policy in other jurisdictions, including South Africa, the euro area, and Australia.



David Archer

David Archer is a former Reserve Bank of New Zealand Assistant Governor (until 2004), including responsibility for monetary policy analysis and advice, forecasting, and research and as a member of the non-statutory monetary policy committee. More recently, he served as Head of the Central Banking Studies Unit and Secretary to the Central Bank Governance Group at the Bank for International Settlements until 2022.⁵⁰



⁵⁰ Mr Archer is a member and trustee of the Reserve Bank Superannuation and Provident Fund Scheme (the Scheme), a retirement income scheme for Reserve Bank staff that closed to new members in the early 1990s. Mr Archer has been involved in past disputes with the Reserve Bank in relation to the Scheme.

Annex 3: How the review was conducted

Consistent with the Terms of Reference, the reviewers considered the written material available to the MPC during the review period, including all MPC papers and supporting material from 2019 to the end of 2022. They also drew on the Reserve Bank's 2022 Review and Assessment of the Formulation and Implementation of Monetary Policy, other published research and reviews, public commentary, historical real-time data, Reserve Bank records, and additional information requests to the Reserve Bank and Treasury.

The reviewers interviewed people who had served on the MPC for at least one year during the review period and, at their discretion, other people who could assist their assessment.

People met with, including virtual meetings (in alphabetical order):

- Alan Bollard
- ANZ Economics team members
- Arthur Grimes
- Former MPC members: Adrian Orr, Bob Buckle, Caroline Saunders, Christian Hawkesby, Geoff Bascand, Peter Harris and Yuong Ha
- Former Treasury observers: Caralee McLiesh, Bryan Chapple and Tim Ng
- Grant Spencer
- International Monetary Fund: Yan Carrière-Swallow, Nada Choueiri, Chris Erceg and Jesper Linde
- John McDermott
- Leo Krippner
- Michael Reddell
- NZIER: Christina Leung and Ting Huang
- Ozer Karagedikli
- Reserve Bank of New Zealand: Anna Breman, Karen Silk, Paul Conway, other staff, and Prasanna Gai
- Suzanne Snively
- The Treasury: Chris Bunny, Mario Di Maio and other staff
- Victoria University of Wellington Macroeconomics Group
- Weshah Razzak
- Westpac Economics team members

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